

2020 Level I Mock Exam (A) PM

The 2020 Level I Chartered Financial Analyst® Mock Examination has 120 questions. To best simulate the exam day experience, candidates are advised to allocate an average of one and a half minutes per question for a total of 180 minutes (3 hours) for this session of the exam.

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- 1 In cases where applicable local laws governing calculation and presentation of investment performance conflict with the GIPS standards, firms are:
- A unable to claim GIPS compliance in cases where local regulations prohibit accurate calculation.
 - B required to calculate and maintain two sets of performance data in order to claim GIPS compliance.
 - C required to comply with local regulations and make full disclosure of the conflict to claim GIPS compliance.

C is correct because in cases where applicable local laws governing calculation and presentation of investment performance conflict with the GIPS standards, firms are required to comply with local regulations and make full disclosure of the conflict in the compliant presentation.

A is incorrect because it is not a requirement for or obstacle to GIPS compliance in cases where local laws conflict with GIPS standards.

B is incorrect because it is not a requirement for or obstacle to GIPS compliance in cases where local laws conflict with GIPS standards.

Global Investment Performance Standards (GIPS)

- 2 After a firm presents a minimum required number of years of GIPS-compliant performance, the firm must present an additional year of performance each year, building up to a minimum of:
- A 10 years of GIPS-compliant performance.
 - B 5 years of GIPS-compliant performance.
 - C 15 years of GIPS-compliant performance.

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A is correct. After a firm presents a minimum of five years of GIPS-compliant performance, the firm must present an additional year of performance each year, building up to a minimum of 10 years of GIPS-compliant performance.

B is incorrect. The initial period is five years of GIPS-compliant performance, building up to a minimum of 10 years of GIPS-compliant performance.

C is incorrect. An additional year of performance each year is added building up to a minimum of 10 years of GIPS-compliant performance.

The GIPS Standards

- 3 According to the GIPS standards, a verification report confirms all of the following **except** whether:
- A specific composite presentations are accurate.
 - B a firm has complied with all firm-wide composite construction requirements.
 - C processes and procedures are designed to calculate and present compliant performance results.



A is correct. According to the Global Investment Performance Standards (GIPS), verification does not ensure the accuracy of any specific composite presentations. Verification tests if a firm has properly constructed composites, and if the firm's systems are designed to properly calculate and present performance in compliance with the GIPS standards. Verification does not, in any way, provide assurance about the results of a specific composite. That level of assurance is provided through an additional level of testing of a specific composite, called a performance examination or performance audit.

B is incorrect because GIPS verification confirms that a firm has complied with all firm-wide composite construction requirements.

C is incorrect because GIPS verification confirms that a firm's processes and procedures are designed to calculate and present compliant performance results.

Introduction to the Global Investment Performance Standards (GIPS)

- 4 Reiko Kimisaki, CFA, is an investment advisor for a national social security fund in a frontier market with a very limited and illiquid capital market. The labor force is young with an investment time horizon of 25 to 30 years. She has been asked to suggest ways to increase the investment return of the overall portfolio. After careful assessment of the fund's previous investment history and available asset classes, she considers investment in private equity. What is Kimisaki's lowest priority to avoid any Code of Ethics and Standards of Professional Conduct violations prior to making this investment recommendation?
- A Assess the risk tolerance of the fund.
 - B Analyze the expected returns of private equity in the market.
 - C Determine if the Investment Policy Statement allows for alternative investments.



B is correct because prior to undertaking analysis with regard to expected returns, an advisor must determine suitability of an investment class including whether it fits within the client's risk tolerance and if it is an allowable asset class as per the client's Investment Policy Statement. Only once these factors have been determined should she proceed if appropriate to analyze expected returns to determine a particular investment recommendation.

A is incorrect because assessing risk of a client is a key role in determining investment suitability.

C is incorrect because before introducing a new asset class, it must be determined if that asset class is an allowable asset class as defined by the Investment Policy Statement.

Guidance for Standards I–VII

- 5 James Simone, CFA, the CFO of a publicly listed company, seeks to improve the quality of his company's communication with institutional fund managers. He holds an investor briefing with this group the evening before the company earnings are announced. The company's quarterly earnings are broadcast in a press release the next day before the market opens. The earnings information in the investor briefing is identical to that in the press release. Did Simone *most likely* violate the CFA Institute Standards of Professional Conduct?

- A Yes
- B No, because investor briefing and press release information are identical.
- C No, because the company releases information while the market is closed.



A is correct because Simone violated Standard II(A)–Material Nonpublic Information by giving institutional fund managers access to material nonpublic information prior to public dissemination (i.e., the press release). By releasing earnings results to a select group of institutional fund managers prior to a public press release, Simone allows the institutional fund managers a time advantage over other investors not invited to the investor briefing.

B is incorrect because it is the timing of the release of the information that causes a violation of Standard II(A).

C is incorrect because despite releasing the information after the close of the market, institutional investors have been given access to nonpublic information in advance of any public dissemination through a press release in violation of Standard II(A).

Guidance for Standards I–VII

- 6 Robin Herring, CFA, is a government bond research analyst at an independent credit rating agency. A competitor credit rating agency just downgraded the bonds of a government Herring follows. Herring notes that all of the information in the competitor's report was covered in his analysis published last week. In the past, Herring has been slow to downgrade bonds, so he starts to doubt his own analysis after seeing the competitor's report. Herring decides to reissue his credit rating of this government bond and match the competitor's downgrade. In his revised report, Herring states that new information has been made available to justify the downgrade. Herring posts the revision on the credit

rating agency's website and provides it by e-mail to all clients who received the original. Herring's rating change least likely violated which of the following CFA Institute Code of Ethics and Standards of Professional Conduct?

- A** Fair Dealing
- B** Communication with Clients
- C** Diligence and Reasonable Basis

A is correct because the analyst has dealt fairly with all clients by sending them an e-mail and posting his rating change on the credit rating agency's website when making material changes to his prior investment recommendation; therefore, he has not violated Standard III(B)–Fair Dealing. Clients should be treated fairly when material changes in a member's or candidate's prior investment recommendations are disseminated, which has been done.

B is incorrect because the analyst has used the release of a competitor's report, contrary to his own previously published report, as a reason to revise his recommendation and has not used reasonable judgment in identifying which factors are important to his investment analyses, recommendations, or actions, as required by Standard V(B)–Communication with Clients and Prospective Clients.

C is incorrect because the analyst does not have a reasonable or adequate basis for his downgrade, as required by Standard V(A)–Diligence and Reasonable Basis. The analyst has also violated Standard I(C)–Misrepresentation which prohibits making misrepresentations relating to investment analysis, recommendations, actions, or other professional activities.

Guidance for Standards I–VII

- 7** Which of the following is *least* likely part of the CFA Institute Standards of Professional Conduct, Standard II–Integrity of Capital Markets? Members and candidates:
- A** must promote the integrity and viability of the global capital markets for the ultimate benefit of society.
 - B** who possess material nonpublic information that could affect the value of an investment must not act or cause others to act on the information.
 - C** must not engage in practices that distort prices or artificially inflate trading volume with the intent to mislead market participants.

A is correct. The Code of Ethics of CFA Institute states that Members of CFA Institute (including CFA charterholders) and candidates for the CFA designation ("Members and Candidates") must promote the integrity and viability of global capital markets for the ultimate benefit of society. It is not part of the CFA Institute Standards of Professional Conduct, Standard II–Integrity of Capital Markets.

B is incorrect. Standard II–Integrity of Capital Markets (A) Material Nonpublic Information states that Members and Candidates who possess material nonpublic information that could affect the value of an investment must not act or cause others to act on the information.

C is incorrect. Standard II–Integrity of Capital Markets (B) Market Manipulation states that Members and Candidates must not engage in practices that distort prices or artificially inflate trading volume with the intent to mislead market participants.

Code of Ethics and Standards of Professional Conduct

- 8 If a firm restructures and wants to remain compliant with the GIPS standards, it should *most likely*:
- A maintain the historical performances for all composites.
 - B alter the historical performances within existing composites.
 - C create all new composites with proper disclosures regarding the reorganization.



A is correct. Changes in a firm's organization must not lead to alteration of historical composite performance.

B is incorrect because changes in a firm's organization must not lead to alteration of historical composite performance.

C is incorrect because changes in a firm's organization must not lead to alteration of historical composite performance.

The GIPS Standards

- 9 Victoria Christchurch, CFA, is a management consultant currently working with a financial services firm interested in curtailing its high staff turnover, particularly amongst CFA charterholders. In recent months, the company lost 5 of its 10 most senior managers, all of whom have cited systemic unethical business practices as the reason for their leaving. To curtail staff turnover by encouraging ethical behavior, it would be *least* appropriate for Christchurch to recommend the company to do which of the following?
- A Implement a whistleblowing policy.
 - B Encourage staff retention with increased benefits.
 - C Create, implement, and monitor a corporate code of ethics.



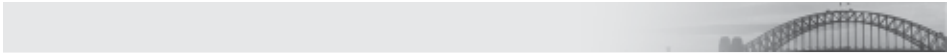
B is correct because the offering of increased benefits to encourage staff retention would not necessarily stop the unethical behavior causing staff turnover and would effectively be asking the ethical employees to ignore the unethical behavior, thus being complicit in the behavior. Under Standard I(A)–Knowledge of the Law, CFA charterholders and candidates must disassociate themselves from unethical behavior. As the unethical business practices are seen as systemic, it would likely require them to leave the firm. Implementing a whistleblowing policy and adopting a corporate code of ethics would likely help to build a foundation of strong ethical behavior.

A is incorrect as introducing a whistleblowing policy would likely help to build a foundation of strong ethical behavior.

C is incorrect as implementing a corporate code of ethics would likely help to build a foundation of strong ethical behavior.

Guidance for Standards I–VII

- 10 When a client asks her how she makes investment decisions, Petra Vogler, CFA, tells the client she uses mosaic theory. According to Vogler, the theory involves analyzing public and nonmaterial nonpublic information including the evaluation of statements made to her by company insiders in one-on-one meetings where management discusses new earnings projections not known to the public. Vogler also gathers general industry information from industry experts she has contacted. Vogler *most likely* violates the CFA Institute Standards of Professional Conduct because of her use of:
- A industry expert information.
 - B one-on-one meeting information.
 - C nonmaterial nonpublic information.



B is correct because a violation of Standard II(A)–Material Nonpublic Information is likely to occur when using information that is selectively disclosed by corporations to a small group of investors, analysts, or other market participants. Earnings estimates given in a one-on-one meeting would likely be considered material and nonpublic information. Information made available to analysts remains nonpublic until it is made available to investors in general. Under the mosaic theory it is acceptable to use information from industry contacts as long as the analyst uses appropriate methods to arrive at her conclusions. Additionally, it is acceptable to use nonmaterial nonpublic information in her analysis, and this use is not a violation of Standard II(A)–Material Nonpublic Information.

A is incorrect because under the mosaic theory it is acceptable to use information from industry contacts as long as the analyst uses appropriate methods to arrive at her conclusions.

C is incorrect because it is acceptable to use nonmaterial nonpublic information in her analysis, and this use is not a violation of Standard II(A)–Material Nonpublic Information.

Guidance for Standards I–VII

- 11 Based on his superior return history, Vijay Gupta, CFA, is interviewed by the First Faithful Church to manage the church's voluntary retirement plan's equity portfolio. Each church staff member chooses whether to opt in or out of the retirement plan according to his or her own investment objectives. The plan trustees tell Gupta that stocks of companies involved in the sale of alcohol, tobacco, gambling, or firearms are not acceptable investments given the objectives and constraints of the portfolio. Gupta tells the trustees he cannot reasonably execute his strategy with these restrictions and that all his other accounts hold shares of companies involved in these businesses because he believes they have the highest alpha. By agreeing to manage the account according to the trustees' wishes, does Gupta violate the CFA Institute Standards of Professional Conduct?
- A No.
 - B Yes, because the manager was hired based upon his previous investment strategy.
 - C Yes, because the restrictions provided by the Trustees are not in the best interest of the members.



A is correct. Standard III(A)–Loyalty, Prudence, and Care, Gupta’s duty of loyalty, prudence, and care is owed to the participants and beneficiaries (members) of the pension plan. As a church plan, the restrictions are appropriate given the objectives and constraints of the portfolio.

B is incorrect because this is irrelevant as the manager has been given a specific mandate by the church trustees.

C is incorrect because the restrictions are appropriate and known to the members as they each individually opt into the plan given the objectives and constraints of the portfolio.

Guidance for Standards I–VII

- 12** Jennifer Ducumon, CFA, is a portfolio manager for high-net-worth individuals at Northeast Investment Bank. Northeast holds a large number of shares in Babyskin Care Inc., a manufacturer of baby care products. Northeast obtained the Babyskin shares when they underwrote the company’s recent IPO. Ducumon has been asked by the investment banking department to recommend Babyskin to her clients, who currently do not hold any shares in their portfolios. Although Ducumon has a favorable opinion of Babyskin, she does not consider the shares a buy at the IPO price nor at current price levels. According to the CFA Institute Code of Ethics and Standards of Professional Conduct the *most* appropriate action for Ducumon is to:

- A** ignore the request.
- B** recommend the shares after additional analysis.
- C** follow the request as soon as the share price declines.



A is correct because Ducumon should refuse to recommend the shares as her opinion of the Babyskin shares must not be affected by internal pressure. If Ducumon followed the request from the investment banking department at her company, she would be in violation of Standard I(B)–Independence and Objectivity. Ducumon must refuse to recommend the Babyskin shares until they are an attractive purchase based on fundamental analysis and market pricing.

B is incorrect because Ducumon should refuse to recommend the shares, as she must issue only recommendations that reflect her independent and objective opinion. Ducumon must refuse to recommend the Babyskin shares until they are an attractive purchase based on fundamental analysis and market pricing.

C is incorrect because Ducumon should refuse to recommend the shares, as she must issue only recommendations that reflect her independent and objective opinion. Ducumon must refuse to recommend the Babyskin shares until they are an attractive purchase based on fundamental analysis and market pricing.

Guidance for Standards I–VII

- 13** Heidi Halvorson, CFA, is the Chief Investment Officer for Tukwila Investors, an asset management firm specializing in fixed-income investments. Tukwila is in danger of losing one of its largest clients, Quinault Jewelers, which accounts for nearly one third of its revenues. Quinault recently told Halvorson that Tukwila would be fired unless the performance of Quinault’s portfolio improves significantly. Shortly after this conversation, Halvorson purchases two corporate

bonds she believes are suitable for any of her clients based upon third party research from a reliable and diligent source. Immediately after the purchase, one bond increases significantly in price while the other bond declines significantly. At the end of the day, Halvorson allocates the profitable bond trade to Quinault and the other bond to two of her largest institutional accounts. Halvorson *most likely* violated the CFA Institute Standards of Professional with regards to:

- A client suitability.
- B trade allocations.
- C third party research.

B is correct because the investment officer failed to deal fairly for her clients by allocating profitable trades to a favored client at the expense of others, a violation of Standard III(B)–Fair Dealing. The standard requires members and candidates to treat all clients fairly when taking investment action. Tukwila should have a systematic approach to allocating trades, such as pro rata, before or at the time of trade execution or as soon as possible after trades are executed.

A is incorrect because the analyst believes the bonds are suitable for any of her clients and has not violated Standard III(C)–Suitability.

C is incorrect because the analyst does have a reasonable or adequate basis for her investment decision, because it is based upon reliable third party research, and has not violated Standard V(A)–Diligence and Reasonable Basis.

Guidance for Standards I–VII

- 14 Manuel Tacqueria, CFA, is a sole proprietor investment adviser managing accounts for a diversified group of clients. Tacqueria obtains his investment research through a subscription service with Alpha Services, a large financial services organization. Tacqueria notes that the research reports are sound because they are extremely detailed and comprehensive. As a result, Tacqueria feels comfortable relying solely upon this research when making recommendations to clients. Tacqueria should *most likely* do which of the following in order to conform to the CFA Institute Code of Ethics and Standards of Professional Conduct?

- A Utilize additional sources of third-party research
- B Undertake and add his own research to the existing reports
- C Conduct additional due diligence on Alpha Services

C is correct because Tacqueria is in violation of Standard V(A)–Diligence and Reasonable Basis as he is required to undertake due diligence efforts on the third-party research provider on a regular basis to ensure that the quality of this research continues to meet his necessary standards.

A is incorrect because the Code and Standards would not require this action. Tacqueria is in violation of Standard V(A)–Diligence and Reasonable Basis as he is required to undertake due diligence efforts on the third-party research provider on a regular basis to ensure that the quality of this research continues to meet his necessary standards.

B is incorrect because the Code and Standards would not require this action. Tacqueria is in violation of Standard V(A)–Diligence and Reasonable Basis as he is required to undertake due diligence efforts on the third-party research provider on a regular basis to ensure that the quality of this research continues to meet his necessary standards.

Guidance for Standards I–VII

- 15 Merchant Capital Partners, a regional investment bank, acts as a market maker for Vital Link Health Services and other small firms listed on an over-the-counter exchange. For those shares for whom Merchant acts as market maker, it trades for its own book as well as engaging in risk arbitrage trading. Merchant allows staff members to trade in shares once clients and the company have traded. Merchant recently obtained material nonpublic information regarding Vital's planned reverse takeover of a publicly listed competitor. In order to be in compliance with the CFA Institute Code and Standards, which type of trading in Vital shares should Merchant *least likely* suspend?
- A Personal
 - B Risk arbitrage
 - C Passive proprietary

C is correct because according to Standard II(A)–Material Nonpublic Information, Recommended Procedures for Compliance, if Merchant stopped market making, a form of proprietary trading, due to being in possession of material nonpublic information, it could tip off investors that Vital is likely to be making a major announcement in the near future. This would be counterproductive to the goals of maintaining the confidentiality of information and providing market liquidity. The Standard recommends that market makers remain passive when in possession of material nonpublic information. The Standard also requires personal trading to be suspended when in possession of material nonpublic information, and it is prudent to suspend arbitrage trading to prevent profits from insider trading.

A is incorrect because when in possession of material nonpublic information, Standard II(A)–Material Nonpublic Information requires personal trading to be suspended.

B is incorrect because when in possession of material nonpublic information, according to Standard II(A)–Material Nonpublic Information, it is prudent to suspend arbitrage trading to prevent profits from insider trading.

Guidance for Standards I–VII

- 16 If you are seeking guidance from the firm's code of ethics or written policies, your actions *most likely* reflect which phase of an ethical decision-making framework?
- A Decide
 - B Reflect
 - C Consider

C is correct. If you are seeking guidance from the firm's code of ethics or written policies, you are in the Consider phase of the ethical decision-making framework. This phase involves taking time to consider the situation influences as well as personal behavioral

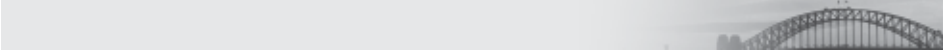
biases that could affect your thinking and decision making. During this phase, you may also seek guidance from such trusted sources as the firm's compliance department or outside counsel.

A is incorrect. The Decide phase of the ethical decision-making framework does not involve seeking guidance from the firm's code of ethics or written policies.

B is incorrect. The Reflect phase of the ethical decision-making framework does not involve seeking guidance from the firm's code of ethics or written policies but rather reflecting on and assessing your decision and its outcome.

Ethics and Trust in the Investment Profession

- 17 Which of the following is *least likely* sufficient to meet recommended or required procedures for compliance with CFA Institute Standard III(A): Loyalty, Prudence, and Care?
- A Disclose any existing conflicts of interest.
 - B Establish a regular client meeting schedule.
 - C Seek best execution when trading on behalf of clients.



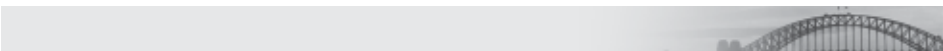
A is correct. Disclosing any existing conflict of interest is least likely adequate to comply with the recommended or required procedure for compliance with CFA Institute Standard III(A): Loyalty, Prudence, and Care. The recommended procedure for compliance states that members and candidates must disclose all actual and potential conflicts of interest so that clients can evaluate those conflicts.

B is incorrect. Establishing a regular client meeting schedule is a recommended or required procedure for compliance with Standard III(A): Loyalty, Prudence, and Care. Members and candidates should establish regularly scheduled client reviews to ensure the investments held adhere to the terms of the clients' governing documents.

C is incorrect. Seeking best execution when trading on behalf of clients, unless directed by the client to do otherwise, is a recommended or required procedure for compliance with Standard III(A): Loyalty, Prudence, and Care.

Guidance for Standards I-VII

- 18 Which of the following statements is *most* accurate? Ethical principles are sets of beliefs centered around:
- A societies' views on what is considered good or bad conduct.
 - B acceptable conduct based on the direct and indirect consequences on others.
 - C rules established by a government with regard to what is acceptable behavior.



A is correct. Ethical principles can be described as societies' beliefs about what is considered good or bad conduct. Ethics can be described as a set of shared beliefs or moral principles regarding standards of behavior expected or required by a community or societal group.

B is incorrect because acceptable conduct based on ethical principles balances self-interest with both the direct and indirect consequences of the behavior for others, not just the direct and indirect consequences alone.

C is incorrect because although governments may create a set of laws they require their citizens to abide by, acceptable behavior considered to be ethical is determined outside any legal requirements. An action can be legal yet unethical.

Ethics and Trust in the Investment Profession

- 19 The probability of Event A is 40%. The probability of Event B is 60%. The joint probability of AB is 40%. The probability (P) that A or B occurs, or both occur, is *closest* to:

- A 40%.
- B 84%.
- C 60%.

C is correct. $P(A \text{ or } B) = P(A) + P(B) - P(AB) = 0.40 + 0.60 - 0.40 = 0.60$ or 60%.

A is incorrect because it uses only the joint probability of 40%.

B is incorrect; it is obtained by adding 0.60 and 0.24 (0.60×0.40) = 0.84.

Probability Concepts

- 20 Assume that a stock's price over the next two periods is as shown below.

Time = 0	Time = 1	Time = 2
$S_0 = 100$	$S_u = 110$	$S_{uu} = 121$
	$S_d = 92$	$S_{ud}, S_{du} = 101.20$
		$S_{dd} = 84.64$

The initial value of the stock is \$100. The probability of an up move in any given period is 40%, and the probability of a down move in any given period is 60%. Using the binomial model, the probability that the stock's price will be \$101.20 at the end of two periods is *closest* to:

- A 48%.
- B 24%.
- C 16%.

A is correct. Across two periods, there are four possibilities: an up move followed by an up move (\$121.00 end value), an up move followed by a down move (\$101.20 end value), a down move followed by an up move (\$101.20 end value), and a down move followed by a down move (\$84.64 end value). The probability of an up move followed by a down move is $0.40 \times 0.60 = 0.24$. The probability of a down move followed by an up move is 0.60×0.40 , which also = 0.24. Both of these sequences result in an end value of \$101.20. Therefore, the probability of an end value of \$101.20 is 48%. Alternatively, the following formula could be used:

$$p(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} = \frac{n!}{(n-x)!x!} p^x (1-p)^{n-x}$$

where

$n = 2$ (number of periods)

$x = 1$ (number of up moves: ud and du)

$p = 0.40$ (probability of an up move)

$$p(1) = \binom{2}{1} 0.40^1 (1 - 0.40)^{2-1} = \frac{2!}{(2-1)!!} \times 0.40^1 \times 0.6^1 = 2 \times 0.40 \times 0.60 = 0.48$$

B is incorrect because it does not recognize that there are two branches that end in \$101.20.

C is incorrect because it is the probability of an up move followed by an up move ($0.4 \times 0.4 = 0.16$).

Common Probability Distributions

- 21** In setting the confidence interval for the population mean of a normal or approximately normal distribution, and given that the sample size is small, Student's t -distribution is the *most* appropriate approach when the variance is:
- A** known.
 - B** large.
 - C** unknown.

C is correct. When the sample size is small (and the population is normally or approximately normally distributed), the Student's t -distribution is preferred if the variance is unknown.

A is incorrect. The z -distribution is preferred if the variance is known.

B is incorrect. When there is a large sample size, not variance, it is possible to use either the z -distribution or t -distribution.

Sampling and Estimation

- 22** A two-tailed t -test of the hypothesis that the population mean differs from zero has a p -value of 0.0275. Using a significance level of 5%, the *most* appropriate conclusion is:
- A** to accept the null hypothesis.
 - B** that the chosen significance level is too high.
 - C** to reject the null hypothesis.

C is correct. The p -value is the smallest level of significance at which the null hypothesis can be rejected. In this case, the given p -value (0.0275) is less than the given level of significance (0.05); therefore, the null hypothesis is rejected.

A is incorrect. Candidates might misunderstand the use of p -value or they may have incorrectly doubled the p -value or inappropriately, in this example, divided the significance level by two.

B is incorrect. The p -value is the smallest level of significance at which the null hypothesis can be rejected. In this case, the given p -value (0.0275) is less than the given level of significance (0.05); therefore, the null hypothesis is rejected.

Hypothesis Testing

- 23** A bank offers an effective annual rate (EAR) of 12%. Assuming quarterly compounding, the stated annual interest rate is *closest* to:
- A** 11.66%.
 - B** 12.55%.
 - C** 11.49%.

C is correct. First, find the periodic interest rate by rearranging the following formula:

$$\text{EAR} = (1 + \text{Periodic interest rate})^m - 1$$

$$12.00\% = (1 + \text{Periodic interest rate})^4 - 1$$

By solving for the periodic interest rate we obtain:

$$\text{Periodic interest rate} = \sqrt[4]{(12.00\% + 1)} - 1 = 2.873734\%$$

Then, find the stated annual interest rate as follows:

$$\begin{aligned} \text{Stated annual interest rate} &= \text{Periodic interest rate} \times m \\ &= 2.873734\% \times 4 \\ &= 11.49\% \end{aligned}$$

A is incorrect. It uses semiannual rather than quarterly compounding:

$$12.00\% = (1 + \text{Periodic interest rate})^2 - 1$$

By solving for the periodic interest rate we obtain:

$$\text{Periodic interest rate} = \sqrt[2]{(12.00\% + 1)} - 1 = 5.83\%$$

Then, find the stated annual interest rate as follows:

$$\text{Stated annual interest rate} = 5.83\% \times 2 = 11.66\%$$

B is incorrect. It uses the following formula:

$$\left(1 + \frac{12.00\%}{4}\right)^4 - 1 = 12.55\%$$

The Time Value of Money

- 24** With Bayes' formula, it is possible to update the probability for an event given some new information. Which of the following *most* accurately represents Bayes' formula?

- A** $P(\text{Event} \mid \text{Information}) = \frac{P(\text{Information} \mid \text{Event})}{P(\text{Information})} P(\text{Event})$
- B** $P(\text{Event} \mid \text{Information}) = \frac{P(\text{Information})}{P(\text{Information} \mid \text{Event})} P(\text{Event})$
- C** $P(\text{Event} \mid \text{Information}) = \frac{P(\text{Information} \mid \text{Event})}{P(\text{Event})} P(\text{Information})$

A is correct. In probability notation, Bayes' formula can be written concisely as

$$P(\text{Event} \mid \text{Information}) = \frac{P(\text{Information} \mid \text{Event})}{P(\text{Information})} P(\text{Event})$$

B is incorrect. In probability notation, Bayes' formula can be written concisely as:

$$P(\text{Event} \mid \text{Information}) = \frac{P(\text{Information} \mid \text{Event})}{P(\text{Information})} P(\text{Event})$$

C is incorrect. In probability notation, Bayes' formula can be written concisely as:

$$P(\text{Event} \mid \text{Information}) = \frac{P(\text{Information} \mid \text{Event})}{P(\text{Information})} P(\text{Event})$$

Probability Concepts

- 25 Using a discount rate of 5%, compounded monthly, the present value (PV) of \$5,000 to be received three years from today is *closest* to:

- A \$4,319.
- B \$4,305.
- C \$4,250.

B is correct. $PV = FV_N (1 + r_s/m)^{-mN}$. In this case, $PV = \$5,000(1 + 0.05/12)^{(-12 \times 3)} = \$4,304.88$.
Using a financial calculator: $FV = \$5,000$, $N = 36$, $I/Y = 5/12$, $PMT = 0$, and solve for PV.

A is incorrect. It is calculated without monthly compounding ($N = 3$, $I/Y = 5$). Alternatively:

$$5,000(1 + 0.05)^{(-3)} = 4,319 \text{ (rounded)}$$

C is incorrect. It is calculated as $PV = 5,000[1 - (0.05 \times 3)] = 4,250$.

The Time Value of Money

- 26 The stated (quoted) annual interest rate on an automobile loan is 10%. The effective annual rate (EAR) of the loan is 10.47%. The frequency of compounding per year for the loan is *closest* to:

- A quarterly.
- B monthly.
- C weekly.

B is correct. $EAR = (1 + \text{Periodic interest rate})^m - 1$.

The solution is found iteratively by substituting the possible frequency of compounding until the EAR is 10.47%.

$$\text{For weekly compounding, } (1 + 0.10/52)^{52} - 1 = 0.10506 = 10.51\%.$$

$$\text{For monthly compounding, } (1 + 0.10/12)^{12} - 1 = 0.10471 = 10.47\%.$$

$$\text{For quarterly compounding, } (1 + 0.10/4)^4 - 1 = 0.10381 = 10.38\%.$$

Thus, the correct answer is monthly compounding.

C is incorrect; $(1 + 0.10/52)^{52} - 1 = 0.10506 = 10.51\%$

A is incorrect; $(1 + 0.10/4)^4 - 1 = 0.10381 = 10.38\%$

The Time Value of Money

- 27 A graphical depiction of a continuous distribution shows the left tail to be longer than the right tail. The distribution is *best* described as having:
- A negative skewness.
 - B leptokurtosis.
 - C positive skewness.

A is correct. A negatively skewed distribution appears as if the left tail has been pulled away from the mean. The average magnitude of negative deviations from the mean is larger than the average magnitude of positive deviations.

B is incorrect. Kurtosis refers to relative peakedness of a distribution; leptokurtosis means more peaked than normal.

C is incorrect. A negatively skewed distribution appears as if the left tail has been pulled away from the mean.

Organizing, Visualizing, and Describing Data

- 28 The returns of a fund are as follows:

Year	Return (%)
1	-20.60
2	15.00
3	0.50
4	9.80
5	4.60

The mean absolute deviation (MAD) of returns for the fund is *closest* to:

- A 13.69%.
- B 9.53%.
- C 11.91%.

B is correct. The mean absolute deviation (MAD) for a sample is calculated as follows:

$$\text{MAD} = \frac{\sum_{i=1}^n |X_i - \bar{X}|}{n}$$

where

X_i = the return of the fund during year i

$\bar{X}$ = the mean of the returns of the sample

n = the number of returns in the sample

i = the index for the year

In this problem:

Mean: $\bar{X} = (-20.60\% + 15.00\% + 0.50\% + 9.80\% + 4.60\%)/5 = 1.86\%$

Deviations from Mean	Absolute Deviation (%)
$ -20.60\% - 1.86\% $	22.46
$ 15.00\% - 1.86\% $	13.14
$ 0.50\% - 1.86\% $	1.36
$ 9.80\% - 1.86\% $	7.94
$ 4.60\% - 1.86\% $	2.74
Total	47.64
MAD = 47.64/5	9.53

C is incorrect. It uses the term " $n - 1$ " in the denominator of the formula of MAD:

$$\frac{\sum_{i=1}^n |X_i - \bar{X}|}{n} = 47.64\%/4 = 11.91\%$$

A is incorrect. It is the sample standard deviation (SD):

$$\begin{aligned} \text{SD} &= \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}} \\ &= \{[(-20.60\% - 1.86\%)^2 + (15.00\% - 1.86\%)^2 + (0.50\% - 1.86\%)^2 + (9.80\% - 1.86\%)^2 + (4.60\% - 1.86\%)^2]/4\}^{0.5} = 13.69\%. \end{aligned}$$

Organizing, Visualizing, and Describing Data

29 Which of the following is *most likely* to be an explanation of the power of a test?

The power of a test is the probability of:

- A rejecting the null when it is false.
- B not accepting the alternative when it is false.
- C a Type I error.

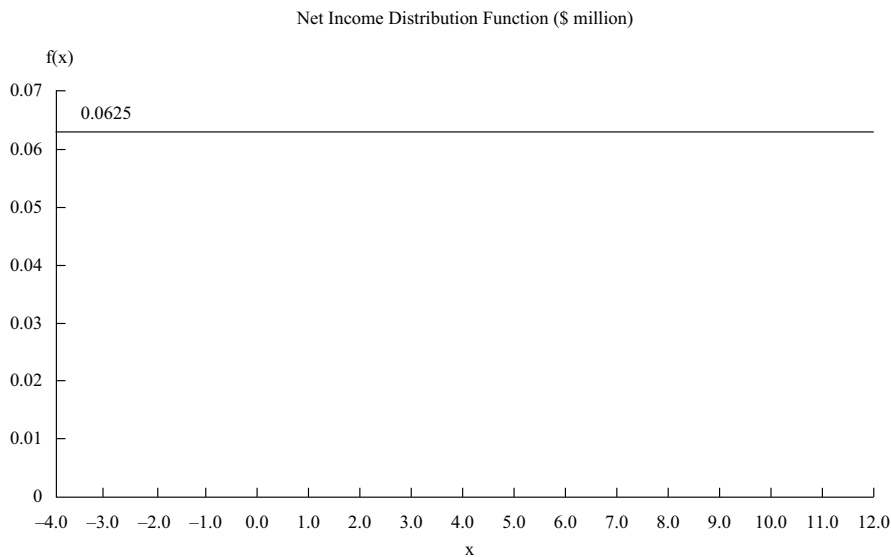
A is correct. The power of a test is the probability of correctly rejecting the null—that is, the probability of rejecting the null when it is false.

B is incorrect. The power of a test is the probability of correctly rejecting the null—that is, the probability of rejecting the null when it is false. If the alternative is not accepted when it is false, it means that a true null hypothesis is not rejected. Therefore, the probability of not accepting the alternative when it is false is not an explanation of the power of a test.

C is incorrect. The power of a test is the probability of correctly rejecting the null—that is, the probability of rejecting the null when it is false. The probability of a Type I error is equal to the probability of rejecting the null when it is true and is also known as the level of significance. Therefore, it is not an explanation of the power of a test.

Hypothesis Testing

- 30** A company forecasts that net income next year will range from a loss of \$4,000,000 to a gain of \$12,000,000. The graph shows the probability density function for this continuous distribution.



Which of the following statements is correct?

- A** The probability of avoiding a loss is 0.25.
- B** The expected net income is \$8,000,000.
- C** The probability of incurring a loss of \$2,000,000 is 0.

C is correct. For a continuous uniform random variable, all outcomes are equally likely, and the probability of the random variable assuming any given fixed value, such as \$2,000,000, is zero.

A is incorrect because the probability of avoiding a loss is the area under the probability density function from zero to \$12,000,000 = $P(0 \leq X \leq 12)$. It includes an area with a base of 12 and a height of $1/16$. Base times height = $12/16 = 75.0\%$ of the total rectangular distribution function.

B is incorrect because, as a continuous uniform random variable, the expected value (mean) for net income is the midpoint between the lowest and highest values:

$$\begin{aligned}
 \mu &= (a + b)/2 \\
 &= (-\$4,000,000 + \$12,000,000)/2 \\
 &= \$8,000,000/2 \\
 &= \$4,000,000
 \end{aligned}$$

Common Probability Distributions

- 31 The market structure in which a firm sells all of the product it produces at the market equilibrium price is *best* described as:
- A oligopoly.
 - B perfect competition.
 - C monopolistic competition.

B is correct. In a perfectly competitive market, sellers have no pricing power and thus sell their product at the price established by demand and supply in the market: the market equilibrium price.

A is incorrect. In an oligopolistic market, there are so few firms in the market that pricing decisions are interdependent.

C is incorrect. In a monopolistically competitive market, sellers have some pricing power as they are able to differentiate their product through advertising or other non-price strategies. Because the product is somewhat different from that of competitors, in the short run the firm can charge the price determined by the demand curve. Unlike perfect competition, there is no well-defined supply function.

The Firm and Market Structures

- 32 With regard to the aggregate demand (AD) curve and an increase in one of its associated factors, which of the following relationships is *least* accurate?

Relationship	Increase in Factor	Shifts the AD Curve	Reason
1	Stock prices	Rightward	Lower investment
2	Consumer confidence	Rightward	Higher consumption
3	Exchange rate*	Leftward	Lower exports and higher imports

* Exchange rate is foreign currency per unit of domestic currency

- A Relationship 3
- B Relationship 1
- C Relationship 2

B is correct. If stock prices rise, the aggregate demand curve will shift to the right (increase in AD) due to higher consumption (wealth effect), not lower investments.

A is incorrect. An increase in foreign exchange rate will make domestic products more expensive, decreasing exports, and imports cheaper, increasing them, causing the AD to shift to left: $(X - M)$ shifts to the left.

C is incorrect. An increase in consumer confidence will cause a shift to the right (AD increase) due to higher consumption.

Aggregate Output, Prices, and Economic Growth

33 The following information is available for 2011:

	New Zealand		Canada	
	Jan 1	Dec 31	Jan 1	Dec 31
Price index	1,137	1,158	117.8	119.9
Nominal exchange rate: NZD/CAD	1.2844			1.2589

The change in the real exchange rate (in NZD/CAD terms) is *closest* to:

- A -2.05%.
- B -1.92%.
- C +1.96%.

A is correct.

	Formula	New Zealand	Canada
Inflation rate	$(I_1 - I_0)/I_0$	$(1,158 - 1,137)/1,137 = 1.85\%$	$(119.9 - 117.8)/117.8 = 1.78\%$
Nominal Exchange Rate Change	$(S_1 - S_0)/S_0$	$(1.2589 - 1.2844)/1.2844 = -1.99\%$	
Real Exchange Rate Change	$\frac{\left(1 + \frac{\Delta S_{\text{NZD/CAD}}}{S_{\text{NZD/CAD}}}\right) \times \left(1 + \frac{\Delta P_{\text{Canada}}}{P_{\text{Canada}}}\right)}{\left(1 + \frac{\Delta P_{\text{NZ}}}{P_{\text{NZ}}}\right)} - 1$		$\frac{(1 - 0.0199) \times (1 + 0.0178)}{(1 + 0.0185)} - 1 = -2.05\%$
Approximation:		$-1.99\% + 1.78\% - 1.85\% = -2.06\%$	

B is incorrect. It uses the correct change in nominal exchange rate, but inverts the inflation rate ratio: $(1 - 0.0199) \times (1.0185)/(1.0178) - 1 = -1.92\%$ or $-1.99\% + 1.85\% - 1.78\%$.

C is incorrect. It uses the appreciation of the Canadian dollar but the correct ratio of inflation rate changes; CAD/NZD: $(0.7943 - 0.7786)/0.7786 = +2.02\%$ or $[1/(1 - 0.0199)] - 1 = 2.03\%$. Giving $(1 + 0.0203) \times 1.0178/1.0185 - 1 = +1.96\%$ or $(+2.03 + 1.78\% - 1.85\%) = +1.96\%$.

Understanding Business Cycles

34 A market structure characterized by homogeneous/standardized product differentiation is *best* described as:

- A perfect competition and oligopoly.
- B monopolistic competition.
- C monopoly.

A is correct. Perfect competition and oligopoly are characterized by homogeneous/standardized product differentiation.

Market Structure	Degree of Product Differentiation
Perfect competition	Homogeneous/standardized
Monopolistic competition	Differentiated
Oligopoly	Homogeneous/standardized
Monopoly	Unique product

B is incorrect because monopolistic competition does not have a standardized product.
C is incorrect because monopolistic competition does not have a standardized product.

The Firm and Market Structures

- 35 Three countries produce tables and chairs, and the output per worker per day in each country as follows:

Country	Tables	Chairs
A	60	80
B	40	60

Assume that Country C produces 10% more tables than Country B and 10% fewer chairs than Country A. Which country *most likely* has the greatest comparative advantage for producing tables? Country

- A C
B B
C A

C is correct. A country has a comparative advantage if its opportunity cost for producing a product is less than the opportunity costs of its trading partners. Notice the cost of a table in units of chairs is lowest for Country A.

Country	Tables	Chairs	Comparative Advantage (Chairs/Tables)
A	60	80	1.33
B	40	60	1.50
C	$40 \times 1.1 = 44$	$80 \times 0.9 = 72$	1.64

B is incorrect because Country A has the lowest ratio of chairs to tables.
A is incorrect because Country A has the lowest ratio of chairs to tables.

International Trade and Capital Flows

- 36 Assume that an economy is composed of two products, X and Y, with the following details:

Product	Quantity Produced in 2012	Quantity Produced in 2013	Product Unit Prices in 2012	Product Unit Prices in 2013
X	351.0	352.0	13.3	13.8
Y	179.0	182.5	unknown	11.1

Assuming 2012 is the base year for measuring GDP and the GDP deflator for the economy in 2013 is 102.4, the unit price of Y in 2012 is *closest* to:

- A 11.2.
- B 10.8.
- C 11.5.

A is correct.

Product	Quantity Produced in 2012 (1)	Quantity Produced in 2013 (2)	Product Unit Prices in 2012 (3)	Product Unit Prices in 2013 (4)	= (2) × (4)
X	351.0	352.0	13.3	13.8	4,857.6
Y	179.0	182.5		11.1	2,025.8
Nominal GDP in 2013 equals the sum of the last column:					6,883.4

$$\begin{aligned}\text{Real GDP}_{2013} &= \text{Nominal GDP}_{2013} \times 100 / \text{GDP deflator} \\ &= 6,883.4 \times 100 / 102.4 = 6,722.1\end{aligned}$$

$$\text{Real GDP}_{2013} = P_{2012}^X \times Q_{2013}^X + P_{2012}^Y \times Q_{2013}^Y$$

$$6,722.1 = (13.3 \times 352.0) + P_{2012}^Y \times 182.5$$

$$6,722.1 = 4,681.6 + P_{2012}^Y \times 182.5$$

$$P_{2012}^Y = 11.2$$

B is incorrect because it applies the GDP deflator to the price of Y in 2013: $(11.1 \times 100) / 102.4 = 10.83$.

C is incorrect because real GDP is calculated with 2012 quantities:

$$6,722.0 = (13.5 \times 351.0) + P_{2012}^Y \times 179.0$$

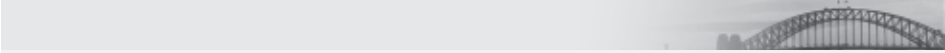
$$6,722.0 = 4,668.3 + P_{2012}^Y \times 179.0$$

$$P_{2012}^Y = 11.50$$

Aggregate Output, Prices, and Economic Growth

37 The primary goal of both monetary and fiscal policy focuses on balancing economic growth and:

- A income distribution.
- B inflation.
- C employment.



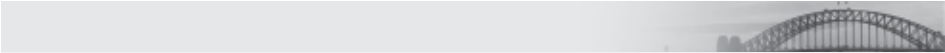
B is correct. The goal of both monetary and fiscal policy is the creation of an economic environment characterized by positive, stable growth and low, stable inflation.

A is incorrect because the distribution of income (and wealth) lies within the purview of fiscal policy involving the government's decisions concerning spending and taxes. By contrast, monetary policy refers to central bank activities directed toward influencing the quantity of money and credit in an economy. Income distribution is not a policy domain of monetary policy.

C is incorrect because the overarching goal of monetary and fiscal policy is to create economic conditions characterized by positive, stable economic growth and low, stable inflation. Achieving this goal promotes stability (rather than cyclical) in employment, consumption, and saving/investment outcomes.

Monetary and Fiscal Policy

- 38** The central bank of a developing country wants to effectively import the inflation experience of the United States. The developing country's economy is *most likely* to experience:
- A** interest rates similar to those in the United States.
 - B** economic growth similar to that of the United States.
 - C** more volatile domestic money supply.



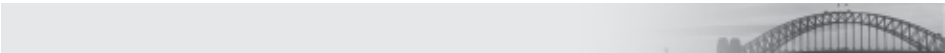
C is correct. When the central bank chooses to target an exchange rate (which effectively imports the developed country's inflation experience), interest rates and conditions in the domestic economy must adapt to accommodate this target and domestic interest rates and money supply can become more volatile.

A is incorrect. Interest rates may become more volatile when the central bank targets the exchange rate rather than the inflation rate.

B is incorrect. The economic growth rate of a developing country is likely to be greater than that of the United States.

Monetary and Fiscal Policy

- 39** If the domestic country desires to reduce a current account surplus, the *most likely* outcome is that it will:
- A** lend to foreign countries.
 - B** encourage foreign direct investment.
 - C** have a lower gross domestic product (GDP) than if it were a closed economy.



A is correct. Because there is a current account surplus, the domestic country will need to lend to foreign countries—most likely, those with current account deficits.

B is incorrect. Foreign direct investment into the domestic country will increase the current account surplus.

C is incorrect. A current account surplus means GDP [$\text{GDP} = \text{Current account surplus} + \text{Private consumption (C)} + \text{Investment (I)} + \text{Government purchases of goods and services (G)}$] is higher than the closed economy version of GDP (i.e., $\text{GDP-closed} = \text{C} + \text{I} + \text{G}$). The difference between the two versions of GDP is positive and equals the current account surplus.

International Trade and Capital Flows

- 40 Which of the following is *most likely* a characteristic of a country that follows the dollarization exchange rate regime? The country:
- A is able to monetize its domestic debt.
 - B has the currency credibility of the US dollar.
 - C has a term structure similar to that of the United States.

B is correct. When a country adopts the US currency as its currency, hence the name *dollarization*, the dollarized country inherits the currency credibility of the US dollar.

A is incorrect. Adopting the US currency will impose fiscal discipline, preventing the government from monetizing its debt.

C is incorrect. The interest rates in a country that adopts the US currency as its currency are generally not the same as in the United States.

Currency Exchange Rates

- 41 A man earns \$3,000 per month and allocates \$300/month for bus travel to visit his children three times per month. The bus company lowers the cost such that he can now take four trips a month for \$320, which he decides to do. His decision to visit his children more often is *most likely* due to which economic effect?
- A Income effect alone
 - B Substitution effect alone
 - C Income and substitution effects combined

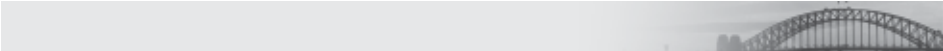
C is correct. The increase in travel is a result of both the income and the substitution effects. His earnings of \$3,000/month have not changed, but because the price has fallen, if he continued to take three trips a month, he would have extra income available. This increase in purchasing power, or real income, is the income effect. The substitution effect is when a consumer buys more of a good when the price falls, as this consumer does by taking four trips instead of three and now allocating more of his budget to bus travel.

A is incorrect. There has been an income effect from the decrease in the price of bus travel, leaving him more income if he continues to take only three trips, but that alone does not account for the man's behavior.

B is incorrect. The substitution effect is when a consumer buys more of a good when the price falls, as this consumer does by taking four trips instead of three and now allocating more of his budget to bus travel. But there is also an income effect from the decrease in the price.

Topics in Demand and Supply Analysis

- 42 When considering the long-run aggregate supply curve, the long run is *best* described as the time required for which of the following combinations of items to become variable?
- A Wages and prices
 - B Wages, prices, and expectations
 - C Wages, prices, expectations, and physical capital

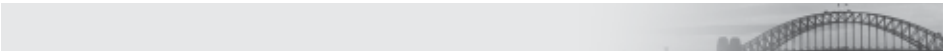


B is correct. The long-run aggregate supply curve represents the level of domestic output that companies will produce at each price level. "Long run" is defined as the time required for wages, prices, and expectations to adjust but not long enough for physical capital to become a variable input: Capital and available technology to use that capital remain fixed.

A and C are incorrect. Wages, prices, and expectations are variable inputs in the long-run aggregate supply curve, but the time is not long enough to allow physical capital to become a variable input.

Aggregate Output, Prices, and Economic Growth

- 43 Which of the following statements is most accurate about the responsibilities of an auditor for a publicly traded firm in the United States? The auditor must:
- A state that the financial statements are prepared according to generally accepted accounting principles.
 - B ensure that the financial statements are free from error, fraud, or illegal acts.
 - C express an opinion about the effectiveness of the company's internal control systems.



C is correct. For a publicly traded firm in the United States, the auditor must express an opinion as to whether the company's internal control system is in accordance with the Public Accounting Oversight Board, under the Sarbanes-Oxley Act. The opinion is given either in a final paragraph in the auditor's report or as a separate opinion.

A is incorrect. The statements are those prepared by management, not the auditor. The auditor is expressing an opinion as to whether the statements are fairly presented and free from material error.

B is incorrect. The auditor only provides reasonable assurance that the statements are free from material error.

Financial Statement Analysis: An Introduction

- 44 For which of the following assets is it *most* appropriate to test for impairment at least annually?
- A Land
 - B A patent with a legal life of 20 years
 - C A trademark with an indefinite expected life



C is correct. Intangible assets with indefinite lives need to be tested for impairment at least annually. Property, plant, and equipment (including land) and intangibles with finite lives are only tested if there has been a significant change or other indication of impairment.

A is incorrect. Property, plant, and equipment (including land) are not tested annually but only when significant events suggest a need to test for impairment.

B is incorrect. Intangible assets with finite lives are not tested annually but only when significant events suggest a need to test for impairment.

Understanding Balance Sheets

- 45 The common shareholders' equity reported on a company's balance sheet is seldom an appropriate measure of the market or intrinsic value of the company's common shares. The *most likely* reason for this fact is that the balance sheet:

- A evaluates a company's financial position spanning a period of time.
- B recognizes items only when future economic benefits are reasonably certain.
- C fails to include all aspects of a company's ability to generate future cash flow.



C is correct. A company's value is a function of many factors, including expected future cash flows and current market conditions. Important aspects of a company's ability to generate future cash flows—for example, its reputation and management skills—are absent from the balance sheet.

B is incorrect. While the statement is true of all financial statements, it is not a reason that intrinsic or market value may be different from equity reported on the balance sheet. The balance sheet is intended to disclose all relevant information about what an entity owns (or controls), what it owes, and what the owners' claims are at a specific point in time.

A is incorrect. The balance sheet information is presented as of a specific point in time. The income statement and cash flow statement are financial statements that are evaluated spanning a period of time.

Understanding Balance Sheets

- 46 Other comprehensive income is *least likely* to include gains or losses on:

- A the sale or disposal of discontinued operations.
- B derivative contracts accounted for as hedges.
- C the translation of foreign currency-denominated subsidiary financial statements.



A is correct. Gains or losses on the disposal of discontinued operations are reported separately near the bottom of the income statement and are included in net income, not other comprehensive income.

B is incorrect. Gains or losses on derivative contracts accounted for as cash flow hedges are included in other comprehensive income.

C is incorrect. Gains or losses on the translation of certain foreign currency-denominated subsidiary financial statements are included in other comprehensive income.

Understanding Income Statements

- 47 The role of the International Organization of Securities Commissions (IOSCO) is *best* described as:
- A promoting cross-border cooperation and uniformity in securities regulation.
 - B enforcing financial reporting requirements for entities participating in capital markets.
 - C promoting the use of International Financial Reporting Standards (IFRS) and the convergence of national accounting standards.

A is correct. IOSCO provides a forum for regulators from different jurisdictions to work together toward fair, efficient, and transparent markets, promoting cross-border cooperation and uniformity in securities regulation.

B is incorrect. This is the role of regulatory authorities such as the Securities Exchange Commission. IOSCO is not a regulator and as such has no authority to regulate.

C is incorrect. This is the role of the IFRS Foundation.

Financial Reporting Standards

- 48 A company uses the straight-line method to depreciate its assets. One of its assets is accounted for under the revaluation model. At the end of Year 1, a revaluation gain is recorded for this asset in other comprehensive income. If there is no further revaluation in Year 2, what is the *most* appropriate depreciable base for the asset in Year 2?
- A No depreciation expense will be recorded under the revaluation model
 - B The asset's value including the revaluation gain
 - C The asset's original cost

B is correct. The revaluation model essentially resets the asset's carrying value to fair value. Depreciation is then calculated based on the new carrying value, which would include the revaluation gain.

A is incorrect. This would be correct under a fair value approach, which is allowed for investment property. No depreciation is recorded under this approach.

C is incorrect. This would be correct under the cost approach.

Long-Lived Assets

- 49 All else being equal and ignoring tax effects, compared with using the straight-line method of depreciation, the use of an accelerated method of depreciation in the early years of an asset's life would *most likely* result in a decrease in the firm's:
- A asset turnover ratio.
 - B shareholders' equity.

- C** cash flow from operations.

B is correct. An accelerated method of depreciation produces greater expenses in the early years and lowers net income, which in turn lowers the retained earnings, resulting in a decrease in shareholders' equity.

A is incorrect. An accelerated depreciation method will lower total net assets in the early years by more than the straight-line method, increasing total asset turnover (for a given sales level).

C is incorrect. CFO is the same under either method. With the accelerated depreciation method the expense is larger, lowering NI, but then the larger expense is added back to the lower NI to yield the same CFO. The accounting policy choice does not affect CFO (ignoring tax effects).

Long-Lived Assets

- 50** The *most* appropriate statement about financial ratio analysis is that it has limited use as an analytical tool for:
- A** providing insights into microeconomic relationships within a company that help analysts project earnings.
 - B** evaluating management.
 - C** comparing companies that use different accounting methods.

C is correct. Financial ratio analysis is limited by the use of alternative accounting methods. Accounting methods play an important role in the interpretation of financial ratios. The lack of consistency across companies makes comparability difficult to analyze and limits the usefulness of ratio analysis.

A is incorrect. Financial ratios provide insights into microeconomic relationships within a company that help analysts project earnings and free cash flow.

B is incorrect. Financial ratios are useful in providing evaluation of management.

Financial Analysis Techniques

- 51** On 1 January 2014, the market rate of interest on a company's bonds is 5%, and it issues a bond with the following characteristics:

Face value	€50 million
Coupon rate, paid annually	4%
Time to maturity	10 years (31 December 2023)
Issue price (per €100)	€92.28

If the company uses International Financial Reporting Standards (IFRS), its interest expense (in millions) in 2014 is *closest* to:

- A** €2.307.
- B** €2.386.
- C** €1.846.

A is correct. IFRS requires the effective interest method for the amortization of bond discounts/premiums. The bond is issued for $0.9228 \times €50 \text{ million} = €46.140$.

$$\text{Interest expense} = \text{Liability value} \times \text{Market rate at issuance} = 0.05 \times €46.140 = €2.307$$

C is incorrect. It uses the coupon rate of $4\% \times 46.140 = 1.8456$.

B is incorrect. It uses the straight-line method of bond amortization, which is allowed under US GAAP but not IFRS.

$$50 \times 4\% - [(100 - 92.28) \times 50]/10 \text{ years} = 2 + 386,000 = 2.386$$

Non-Current (Long-Term) Liabilities

- 52 Information about a company's historical performance for the last two years and additional information are summarized in the following table.

(\$ thousands)	2013	2012
Sales	5,500.0	5,350.0
Cost of goods sold	-2,200.0	-2,140.0
Operating expenses	-2,350.0	-2,350.0
Gain on sale of short-term investments	0	140.0
Tax expense	-237.5	-325.0
Income (loss) from discontinued operations (net of tax)	-312.5	112.5
Net income	400.0	787.5

Industry sales are expected to increase 5%, and the company expects to maintain its current market share and gross profit margin. Operating expenses are not expected to change with the increase in sales.

The company sold off its portfolio of marketable securities in 2012 and used the funds to purchase operating assets. In 2012, the company announced its intention to sell off a division, and that sale was completed in 2013. The results from the division and the gain or loss incurred on the sale are classified as discontinued operations.

The projected net income (in thousands) for 2014 is *closest* to:

- A \$745.
- B \$836.
- C \$635.

B is correct. The loss (gain) from discontinued operations and the gain on the sale of the portfolio investments should not be included in the forecast because they are not recurring items. First, the recurring operating margin before tax should be forecasted, noting that the operating costs are fixed costs, and then the tax rate from 2013 should be used to determine net income.

(\$ thousands)	2014 forecast	2013		2012	
Sales (increase 5%)	\$5,775	\$5,500.0	100%	\$5,350.0	100%
Cost of goods sold (40% each year)	2,310	2,200.0	40%	2,140.0	40%
Operating expenses (fixed cost)	2,350	2,350.0		2,350.0	
Recurring operating income	1,115	950.0		860	
Tax expense (25% × operating income)	279	237.5	(237.5/950) = 25%		
Net income	\$836				

A is incorrect. It adjusts for the discontinued operations before calculating the margins

(\$ thousands)	2014 Forecast	2013	2012
Sales	5,775	5,500	5,350
Net income as reported		\$400	\$787.5
Discontinued operations loss (gain)		312.5	(112.5)
Recurring EAT		712.5	675
Recurring EAT margin		12.9%	12.6%

Using the EAT margin of 12.9% × 5,775 = 745

C is incorrect. It takes the average margin of the two previous years without any adjustments for the non-recurring items or change in tax rate. (400/5,500 = 7.3%) and (787.5/5,350 = 14.7%).

$$\frac{1}{2}[7.3\% + 14.7\%] = 11.0\%$$

$$0.11 \times 1.05 \times 5,500 = \$635$$

Understanding Income Statements

- 53 Which of the following statements *most* accurately describes a valuation allowance for deferred taxes? A valuation allowance is required under:
- A both IFRS and US GAAP on deferred tax assets arising from the translation of foreign operations.
 - B IFRS on revaluation of a deferred tax asset.
 - C US GAAP if there is doubt about recovering a deferred tax asset.

C is correct. A valuation allowance is required under US GAAP if there is doubt about whether a deferred tax asset will be recovered. Under IFRS, the deferred tax asset is written down directly.

A is incorrect. Tax differences arising from foreign operations currency adjustments go to equity. There are no deferred tax assets.

B is incorrect. IFRS does not have the concept of valuation allowance for deferred tax assets. Deferred tax assets are written down directly instead.

Income Taxes

- 54 A company has operated at full capacity throughout the year, and a review of its inventory records for that period indicate that the following costs were incurred:

Fixed production overhead	\$500,000
Direct material and direct labor	\$300,000
Storage costs incurred during production	\$25,000
Abnormal waste costs	\$30,000

The total capitalized costs to inventory during the year are *closest* to:

- A \$800,000.
 B \$855,000.
 C \$825,000.

C is correct. The total capitalized costs include fixed production costs, the direct conversion costs of material and labor, and storage costs required as part of production. They do not, however, include abnormal waste costs.

$$\$500,000 + \$300,000 + \$25,000 = \$825,000$$

A is incorrect. It does not include the storage costs during production.

B is incorrect. It includes the abnormal waste cost.

Inventories

- 55 A company purchased equipment for \$50,000 on 1 January 2011. It is depreciating the equipment over a period of 10 years on a straight-line basis for accounting purposes, but for tax purposes it is using the declining balance method at a rate of 20%. Given a tax rate of 30%, the deferred tax liability at the end of 2013 is *closest* to:

- A \$6,720.
 B \$2,820.
 C \$420.

B is correct. The deferred tax liability is equal to the tax rate $\times$ temporary difference between the carrying amount of the asset and the tax base.

Value for accounting purposes after three years	$50,000 - [3 \times (50,000/10)] =$	\$35,000
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Value for tax purposes:

$$\text{Carrying amount} = \text{Start of year balance} \times (1 - 0.20)$$

After three years:	$50,000 \times 0.8 \times 0.8 \times 0.8 =$	<u>25,600</u>
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Temporary difference		9,400
Deferred tax liability at 30%:	$30\% \times 9,400 =$	\$2,820

A is incorrect. It is the sum of the three years of liabilities (1,500 + 2,400 + 2,820), but these amounts are ending liability amounts, not a cumulative amount.

C is incorrect. It is the deferred tax expense for the year, the difference between 2012 and 2013: $2,820 - 2,400 = 420$.

Income Taxes

- 56 The following information was taken from the financial statements and notes to the financial statements of a company that accounts for its inventory on a LIFO basis.

**For Periods Ending December 31,
in \$ thousands**

	2014	2013
Net sales	\$11,159	\$8,895
Net income	178	81
Inventories (LIFO basis)	1,406	2,220
Total assets	5,570	6,288
LIFO reserve	\$867	\$547

The company's tax rate for the current and all prior years is 33.3%.

If the company had reported on a FIFO basis and the additional tax liability arising from restatement resulted in an immediate reduction in cash, its 2014 return on assets (using 2014 assets only) would be:

- A 6.2%.
- B 6.4%.
- C 7.7%.

B is correct. The net income and total assets both need to be adjusted to what they would have been under the FIFO method.

$$\begin{aligned}
 \text{Return on assets} &= (\text{Net income} / \text{Total assets}) \times 100 \\
 &= (391.4 / 6,148.3) \times 100 \\
 &= 6.36\%
 \end{aligned}$$

Under FIFO, total assets increase by the LIFO reserve but decrease by the cash paid for the cumulative amount of additional income taxes that would arise. Net income will be higher under FIFO as a result of lower COGS—that is, the increase in the LIFO reserve—but will be reduced by the taxes paid on the increase in operating profit.

(\$ thousands)

LIFO net income	178	
+ Reduction in COGS	+320	Increase in LIFO reserve in 2014: $867 - 547$
– Tax on increased operating profit	–106.6	$33.3\% \times 320$
FIFO Net income	391.4	

(continued)

(\$ thousands)

LIFO Total assets	5,570	
+ FIFO Increase in inventory	867	Add LIFO reserve: 867
– Tax paid on higher cumulative profits	–288.7	33.3% × 867*
FIFO Total assets	6,148.3	

* Cumulative tax saving: 2014 tax rate for total LIFO reserve

A is incorrect. It uses the tax for only the current year, 2014, not the cumulative tax savings in the asset determination:

$$\text{ROA} = [391.4 / (5,570 + 867 - 106.6)] \times 100 = (391.4 / 6,330) \times 100 = 6.18\%$$

C is incorrect. It ignores tax effects but adjusts pre-tax income and inventory properly:

$$\text{ROA} = [(178 + 320) / (5,570 + 867)] \times 100 = (498 / 6,437) \times 100 = 7.7\%$$

Inventories

- 57 Which of the following conditions is *most likely* associated with decreased earnings quality? Compared with the prior year, the reporting entity's earnings:
- A decreased slightly in response to the introduction of conservative accounting policies.
 - B were similar in magnitude but included a large gain on the sale of a manufacturing plant.
 - C increased slightly because of a reduction in bad debt expense based on more-current experiences.

B is correct. The sale of a manufacturing plant is likely a one-time transaction that will not be sustained in future years. The quality of reported earnings has therefore decreased from the prior year.

A is incorrect. This is an example of decreased financial reporting quality because conservatism, a choice made by management, is making it more difficult to establish expectations for the future. Since the earnings only decreased because of the conservatism, there is no decrease in the underlying earnings quality.

C is incorrect. If the estimates are based on more recent experiences, it does not imply the intent to manipulate earnings and will provide a more faithful representation of the company's performance.

Financial Reporting Quality

- 58 A credit analyst is *most likely* to place more focus on:
- A operating leverage than financial leverage.
 - B cash flows than accrual income.
 - C upside potential than downside risk.

B is correct. Credit analysts are particularly focused on assessing debt-paying ability, which is generated from cash flows and not from accrual-based measures, such as net income. The return to debtholders is limited by contract, so upside potential is less important than concern for loss.

Financial leverage is a major concern for credit analysts because it is associated with a greater chance of default. Operating leverage, which measures the volatility of operating income as a result of fixed costs, is a lesser concern.

A is incorrect. Credit analysts are more concerned with financial leverage because higher financial leverage is associated with higher levels of default.

C is incorrect. Debt providers are more concerned with downside risk because they are entitled only to contractual cash flows. Additional upside potential does not result in additional return.

Financial Statement Analysis: Applications

- 59 When the market rate of interest falls after issuance, a company selecting the fair value option for reporting a liability with a fixed coupon rate will report:
- A no change.
 - B a gain.
 - C a loss.

C is correct. A company selecting the fair value option for a liability with a fixed coupon rate will report a loss when market interest rates decrease.

A is incorrect because a company selecting the fair value option for a liability with a fixed coupon rate will report losses when market interest rates decrease.

B is incorrect because a company selecting the fair value option for a liability with a fixed coupon rate will report losses (not gains) when market interest rates decrease.

Non-Current (Long-Term) Liabilities

- 60 An analyst gathers the following information from a company's current financial statements:

Year Ended 31 December (\$ millions)	
	2016
Revenue	26,430
Cost of goods sold	12,831
Operating expenses	9,802
Income tax expense	1,277
Net Income	2,250

Year Ended 31 December (\$ millions)

	2016	2015
Accounts receivable	1,134	1,072
Accounts payable	4,858	3,724
Inventory	4,462	3,670

If the company uses the direct method to prepare its cash flow statement, the cash received from customers (in \$ millions) will be *closest* to:

- A 26,368.
- B 25,296.
- C 26,492.

A is correct. Accounts receivable increased; therefore, cash collected from customers will be less than reported revenue. Calculations are as follows:

	Calculation	\$ Millions
Revenue		26,430
Less increase accounts receivable	$1,134 - 1,072 =$	62
Cash received from customers		26,368

B is incorrect. This answer just subtracts ending A/R. They understand A/R aren't collected but does not consider opening A/R. $26,430 - 1,134 = 25,296$. OR they subtract the increase in payables which we jiggered to $= 1,134$.

C is incorrect. This answer mistakenly adds the change in receivables to revenue instead of deducting it.

Revenue	26,430
PLUS increase accounts receivable	$1,134 - 1,072 =$ 62
Cash received from customers	26,492

Understanding Cash Flow Statements

61 The following information is available for a firm:

Market risk premium	7.0%
Risk-free rate	2.0%
Comparable firm return	10.4%
Comparable firm debt-to-equity ratio	1.0
Comparable firm tax rate	40.0%

The firm's unleveraged beta is *closest* to:

- A 0.75.
- B 1.20.
- C 1.05.

A is correct. Find the comparable firm's beta: $(10.4\% - 2.0\%)/7.0\% = 1.20$.

Unlever the comparable firm's beta: $\beta_{L, \text{comparable}}/[1 + (1 - \text{Tax rate}) \times \text{Debt-to-equity ratio}]$

$$1.20/[1 + (1 - 40\%) \times 1.0] = 0.75.$$

B is incorrect because it is the comparable firm's beta.

C is incorrect because the comparable firm's beta is computed as $1.68 = (10.4\% - 2.0\%)/(7.0\% - 2.0\%)$ and then proceeds to the next calculation.

Cost of Capital

- 62** A company has an equity beta of 1.4 and is 60% funded with debt. Assuming a tax rate of 35%, the company's asset beta is *closest* to:

- A** 0.98.
B 1.01.
C 0.71.

C is correct. Note: 60% debt financing is equivalent to a debt-to-equity ratio of $1.50 = 0.60/(1 - 0.60)$.

$$\begin{aligned}\beta_{\text{Asset}} &= \beta_{\text{EQ}} \times \left\{ 1/[1 + (1 - t)D/E] \right\} \\ &= 1.4/[1 + (1 - 0.35) \times 1.5] \\ &= 0.7089\end{aligned}$$

A is incorrect because it uses the inverse of the D/E ratio:

$$0.9766 = 1.4/[1 + (1 - 0.35)66.7\%]$$

B is incorrect because it uses the 60% debt financing instead of the D/E ratio:

$$1.0072 = 1.4/[1 + (1 - 0.35)60\%]$$

Cost of Capital

- 63** The following information is available for a firm:

Revenue	£800,000
Variable cost	400,000
Fixed cost	200,000
Operating income	200,000
Interest	60,000
Net income	140,000

The firm's degree of total leverage (DTL) is *closest* to:

- A** 1.43.
B 2.00.
C 2.86.

C is correct. $DTL = \text{Revenue} - \text{Variable cost} / \text{Net income} = £800,000 - £400,000 / £140,000 = 2.86$.

A is incorrect because it is the degree of financial leverage (DFL).

$DFL = \text{Operating income} / \text{Net income} = £200,000 / £140,000 = 1.43$

B is incorrect because it is the degree of operating leverage (DOL).

$DOL = (\text{Revenue} - \text{Variable cost}) / \text{Operating Income} = [(£800,000 - £400,000) / £200,000] = 2.00$

Measures of Leverage

64 An inventory system that reduces average inventory without affecting sales will *most likely* reduce the:

- A** quick ratio.
- B** inventory turnover.
- C** cash conversion cycle.

C is correct. A reduction in inventory will increase the inventory turnover (Cost of goods sold/Average inventory), which means that the days in inventory will be reduced ($365 / \text{Inventory turnover}$). This will lead to a reduction in the cash conversion cycle (also called net operating cycle). Cash conversion cycle consists of number of days of inventory and number of days of receivables minus number of days of payables.

A is incorrect. Quick ratio will increase as the result of efficient inventory system.

B is incorrect. Inventory turnover will increase as the result of efficient inventory system.

Financial Analysis Techniques

65 The post-audit performed as part of the capital budgeting process is *least likely* to include the:

- A** provision of future investment ideas.
- B** rescheduling and prioritizing of projects.
- C** indication of systematic errors.

B is correct. Rescheduling and prioritizing projects is part of the planning stage of the capital budgeting process, not the post-audit. The post-audit's purpose is to explain any differences between the actual and predicted results of a capital budgeting project. This process can aid in indicating systematic errors, improve business operations, and provide concrete ideas for future investment opportunities.

A is incorrect. Rescheduling and prioritizing projects is part of the planning stage of the capital budgeting process, not the post-audit. The post-audit's purpose is to explain any differences between the actual and predicted results of a capital budgeting project. This process can aid in indicating systematic errors, improve business operations, and provide concrete ideas for future investment opportunities.

C is incorrect. Rescheduling and prioritizing projects is part of the planning stage of the capital budgeting process, not the post-audit. The post-audit's purpose is to explain any differences between the actual and predicted results of a capital budgeting project. This process can aid in indicating systematic errors, improve business operations, and provide concrete ideas for future investment opportunities.

Capital Budgeting

- 66** Which of the following statements describes the *most* appropriate treatment of cash flows in capital budgeting?
- A** Interest costs are included in the project's cash flows to reflect financing costs.
 - B** A project is evaluated using its incremental cash flows on an after-tax basis.
 - C** Sunk costs and externalities should not be included in the cash flow estimates.



B is correct. All of the incremental cash flows arising from a project should be analyzed on an after-tax basis.

C is incorrect. Only sunk costs should be ignored in a project's cash flow estimation, but not any externalities. Sunk costs cannot be recovered once they have been incurred. Externalities (both positive and negative ones) are the effects of an investment decision on other things beside the investment itself; they should therefore be included in the cash flow estimation.

A is incorrect. Financing costs like interest costs are excluded from calculations of operating cash flows. The financing costs are reflected in the required rate of return for an investment project. If financing costs are included, we would be double-counting these costs.

Capital Budgeting

- 67** In order to maintain an adequate net daily cash position, a company is *least likely* to:
- A** monitor access to borrowing facilities.
 - B** forecast depreciation and accruals.
 - C** predict the business cycles and seasonal effects.



B is correct. Accruals are paid at a later date, and depreciation is a noncash expense.

A is incorrect. Most companies maintain a cash buffer as a protection from unexpected cash need or to provide the financial flexibility. The size of the buffer depends on several influences including the company's ability to access other liquidity sources or borrowing facilities.

C is incorrect. Correctly predicting the peak need caused by seasonality or other non-operating activities such as M&A or major capital expenditures can help the company to forecast cash required to cover the need.

Working Capital Management

- 68 Assume a 365-day year and the following information for a company:

	Current Year	Previous Year
Sales	\$12,000	\$10,000
Cost of goods sold	\$9,000	\$7,500
Inventory	\$1,200	\$1,000
Accounts payable	\$600	\$600

The firm's days of payables for the current year is *closest* to:

- A 23.8.
B 18.3.
C 24.9.

A is correct. The number of days of payables is

$$\begin{aligned}\frac{\text{Accounts payable}}{(\text{Purchases}/365)} &= \frac{\text{Accounts payable}}{[(\Delta \text{ inventory} + \text{COGS})/365]} \\ &= \frac{\$600}{[(\$1,200 - \$1,000 + \$9,000)/365]} \\ &= 23.8\end{aligned}$$

B is incorrect because it uses sales instead of cost of goods sold and ignores the change in inventory: $600/(12,000/365) = 18.3$.

C is incorrect because it miscalculates purchases: $600/[(1,000 - 1,200 + 9,000)/365] = 24.9$.

Financial Analysis Techniques

- 69 A 30-day \$10,000 US Treasury bill sells for \$9,932.40. The discount basis yield (DBY) is *closest* to:

- A 8.11%.
B 8.17%.
C 8.28%.

A is correct.

$$\begin{aligned}\text{DBY} &= \frac{\text{Face value} - \text{Purchase price}}{\text{Face value}} \times \frac{360}{\text{Days to maturity}} \\ &= \frac{\$10,000 - \$9,932.40}{\$10,000} \times \frac{360}{30} \\ &= 8.11\%\end{aligned}$$

B is incorrect because it is the money market yield.

$$\begin{aligned}\text{MMY} &= \frac{\$10,000 - \$9,932.40}{\$9,932.40} \times \frac{360}{30} \\ &= 8.167\%\end{aligned}$$

C is incorrect because it is the bond equivalent yield.

$$\begin{aligned}\text{BEY} &= \frac{\$10,000 - \$9,932.40}{\$9,932.40} \times \frac{365}{30} \\ &= 8.281\%\end{aligned}$$

Working Capital Management

70 If a 90-day \$10,000 US Treasury security is selling for \$9,870, the discount-basis yield is *closest* to:

- A** 5.27%.
- B** 5.34%.
- C** 5.20%.

C is correct.

$$\begin{aligned}\text{DBY} &= \left(\frac{\text{Face value} - \text{Purchase price}}{\text{Face value}} \right) \times \left(\frac{360}{\text{Days to maturity}} \right) \\ &= \left(\frac{\$10,000 - \$9,870}{\$10,000} \right) \times \left(\frac{360}{90} \right) \\ &= 0.0520\end{aligned}$$

A is incorrect because it is the money market yield [= discount basis yield × (\$10,000/\$9,870)].

B is incorrect because it is the bond equivalent yield [= discount basis yield × (\$10,000/\$9,870) × (365/360)].

Working Capital Management

71 Based on best practices in corporate governance procedures, it is *most* appropriate for a company's compensation committee to:

- A** link compensation with long-term objectives.
- B** include a retired executive from the firm.
- C** include a representative from the firm's external auditor.

A is correct. Under appropriate corporate governance procedures, the compensation committee should link compensation with long-term objectives.

B is incorrect because the committee should be composed of independent members only. Good corporate governance procedures would require that executive (internal) directors not rule on matters underlying conflicts of interest or on matters requiring an unbiased judgment (such as audit, remuneration, or related-party transaction matters). Retired executives and external auditors are not independent and should not be a part of the compensation committee.

C is incorrect because the committee should be composed of independent board members only. Good corporate governance procedures would require that executive (internal) directors not rule on matters underlying conflicts of interest or on matters

requiring an unbiased judgment (such as audit, remuneration, or related-party transaction matters). Retired executives and external auditors are not independent and should not be a part of the compensation committee.

Corporate Governance and ESG: An Introduction

- 72 Which of the following is *most likely* associated with poor corporate governance?
- A Reduction in exposure to regulatory actions
 - B Increased control and compliance monitoring of corporate decisions
 - C Management of a company to a lower risk profile relative to shareholder tolerance

C is correct. Poor governance can result in ineffective decision making. Management may make decisions that benefit themselves at the cost of shareholders, such as taking less risk than is appropriate to create a more stable environment. Managing the company at a lower risk profile than necessary based on shareholders' tolerance is an example of ineffective decision making. Increased control and compliance monitoring are usually the result of good corporate governance. A reduction in exposure to regulatory actions would be the result of good, not poor, corporate governance.

A is incorrect. A reduction in exposure to regulatory actions would be the result of good, not poor, corporate governance.

B is incorrect. Increased control and compliance monitoring are usually the result of good corporate governance.

Corporate Governance and ESG: An Introduction

- 73 If the following three stocks are held in a portfolio, the portfolio's total return on an equal-weighted basis is *closest* to:

Stock	Number of Shares Owned	Beginning of Period Price per Share (\$)	End of Period Price per Share (\$)	Dividend per Share during the Period (\$)
A	500	40	37	2.00
B	320	50	52	1.50
C	800	30	34	0.00

A 3.28%.

B 5.94%.

C 6.37%.

B is correct. Equal weighting assigns an equal weight to each constituent security at inception. Therefore, it is the sum of the total return from each security divided by the number of securities in the portfolios.

Stock	$(P_1 - P_0 + D)/P_0$	Total Return (%)
A	$(37 - 40 + 2.00)/40 =$	-2.5
B	$(52 - 50 + 1.50)/50 =$	7.00
C	$(34 - 30 + 0)/30 =$	13.33
Portfolio return with equal weighting: $(-2.50 + 7.00 + 13.33)/3 =$		5.94

A is incorrect. It is computed on a price return basis not total return.

Stock	Price Return (%) = $(P_1 - P_0)/P_0$
A	$(37 - 40)/40 = -7.50$
B	$(52 - 50)/50 = 4.00$
C	$(34 - 30)/30 = 13.33$
Portfolio return with equal weighting: $(-7.50 + 4.00 + 13.33)/3 = 3.28\%$	

C is incorrect. It is the total return on the basis of beginning of period market-capitalization weights.

Stock	Total Return (%) = $(P_1 - P_0 + D)/P_0$	BOP Weights*	Total Return (%) × BOP Weight
A	$(37 - 40 + 2.00)/40 = -2.50$	0.333	-0.83
B	$(52 - 50 + 1.50)/50 = 7.00$	0.267	1.87
C	$(34 - 30 + 0)/30 = 13.33$	0.400	5.33
Portfolio market-capitalization-weighted total return =			6.37

* BOP weights:

Beginning Value of Portfolio: $(A = 500 \times \$40) + (B = 320 \times \$50) + (C = 800 \times \$30) = \$60,000$;
BOP weights: $A = 20,000/60,000 = 0.333$; $B = 16,000/60,000 = 0.267$; $C = 24,000/60,000 = 0.400$

Security Market Indexes

74 When parties exchange fixed cash payments for payments that depend on the returns to a stock or a stock index, they are purchasing a(n):

- A equity swap.
- B index fund.
- C stock option.

A is correct. Equity swaps consist of parties exchanging fixed cash payments for payments that depend on the returns to a stock or a stock index.

B is incorrect. The payments depend on the returns to a stock or a stock index, but an index fund has not been directly purchased.

C is incorrect. An option contract allows the holder (the purchaser) of the option to buy or sell an underlying instrument at a specified price at or before a specified date in the future.

Market Organization and Structure

75 Dark pools are *best* described as:

- A trading venues that exercise little regulatory authority over their subscribers.
- B operated by investment dealers that specialize in high-risk securities.
- C certain groups of similar assets that issue securities representing shared ownership.

A is correct. Dark pools are trading venues that function like exchanges but do not exercise regulatory authority over their subscribers except with respect to the conduct of their trading in those venues.

B is incorrect. Dark pools may be operated by investment dealers but do not necessarily trade high risk securities.

C is incorrect. This is the definition of pooled investment vehicles

Market Organization and Structure

76 A security market index that reports returns based on the reinvestment of income and the change in price of its constituent securities is *best* described as which type of index?

- A Total return
- B Float-adjusted
- C Price return

A is correct. A total return index reflects not only the constituent securities' prices but also the reinvestment of all income since inception.

B is incorrect. Float-adjustment is related to the weighting of constituent securities within the index and is not related to whether the index is reported based on price return or total return.

C is incorrect. A price index reflects only the prices of the constituent securities within the index and excludes the reinvestment of income.

Security Market Indexes

77 When constructing a list of peer companies to be used in equity valuation, which of the following would *least likely* improve the group? Companies in the same peer group should ideally:

- A be exposed to similar stages in the business cycle.
- B have similar valuations.
- C have the effects of finance subsidiaries minimized.

B is correct. Companies in the same peer group can have different valuations depending on structure and competitiveness.

A is incorrect. Valuations may be of limited value when comparing companies that are exposed to different stages of the business cycle.

C is incorrect. To make a meaningful comparison of companies, analysts should make adjustments to the financial statements to lessen the impact that the finance subsidiaries have on the various financial metrics being compared.

Introduction to Industry and Company Analysis

- 78** A company has issued only one class of common shares, and it does not pay dividends on them. It has also issued two types of non-cumulative preference shares: one that is putable and the other callable. Which of these securities will *most likely* offer the lowest expected return to the investor?
- A** Putable preference shares
 - B** Common shares
 - C** Callable preference shares



A is correct. Putable preference shares are less risky than their callable counterparts. They give the investor the option to put the shares back to the company. Because of the lower risk, they will provide a lower expected rate of return. Common shares are the most risky, whether or not they are dividend paying, and are likely to offer the highest expected return.

B is incorrect. Common shares are the most risky, whether or not dividend paying, and are likely to offer the highest expected return. Though these preference shares have non-cumulative dividend feature, preference shareholders will receive priority if the company is liquidated.

C is incorrect. Callable preference shares are more risky than their putable counterparts and so they are likely to offer higher expected returns than putable preference shares.

Overview of Equity Securities

- 79** An industry characterized by rising volumes, improving profitability, falling prices, and relatively low competition among companies is *most likely* in which of the following life-cycle stages?
- A** Growth
 - B** Mature
 - C** Embryonic



A is correct. An industry in growth stage is characterized by rising volumes, improving profitability, falling prices, and relatively low competition among companies.

B is incorrect. In the mature stage there will be little or no growth and relatively stable demand for products.

C is incorrect. In the embryonic stage there will be slowing growth and high prices.

Introduction to Industry and Company Analysis

- 80** The following market information relates to a company:

Market price per share	\$37.80
Number of shares outstanding	1,000,000
Net income	\$5,250,000
Total common equity	\$35,000,000
Total annual dividend paid	\$1,512,000
Risk-free rate	2.60%
Market risk premium	8.00%
Beta	1.05

Using the capital asset pricing model (CAPM), the company's cost of equity is closest to:

- A 15.0%.
- B 12.4%.
- C 11.0%.

C is correct. Using the CAPM:

$$\begin{aligned}\text{Cost of equity} &= \text{Risk-free rate} + (\text{Beta} \times \text{Market risk premium}) \\ &= 2.6 + (1.05 \times 8) = 11\%\end{aligned}$$

B is incorrect. It is the Dividend yield + (Beta × Market risk premium)

$$\text{Dividend per share} = 1,512,000 / 1,000,000 = 1.512$$

$$\text{Dividend yield} = 1.512 / 37.8 = 4\%$$

$$\text{Cost of equity (with the mistake)} = 4\% + (1.05 \times 8) = 12.4\%$$

A is incorrect. It is the Return on equity = Net income/Common equity

$$\begin{aligned}\text{ROE} &= 5,250,000 / 35,000,000 \\ &= 15\%\end{aligned}$$

Overview of Equity Securities

81 An investor has gathered the following data for a common stock.

Earnings per share, 2013	\$2.50
Dividend payout ratio, 2013	60%
Dividend growth rate expected during 2014 and 2015	25%
Dividend growth rate expected after 2015	5%
Investors' required rate of return	12%

Using the two-stage dividend discount model, the value per share of this common stock is *closest* to:

- A \$38.70.
- B \$31.57.
- C \$28.57.

B is correct. Dividend per share (2013) = $\$2.50(0.6) = \1.50 .

$$V = \frac{1.50(1.25)}{1.12} + \frac{1.50(1.25)^2}{1.12^2} + \frac{1.50(1.25)^2(1.05)}{(0.12 - 0.05)} \times \frac{1}{1.12^2}$$

$$= \$1.67 + \$1.87 + \$28.03 = \$31.57$$

C is incorrect. The terminal value is discounted for 3 years instead of 2 years.

$$V = \frac{1.50(1.25)}{1.12} + \frac{1.50(1.25)^2}{1.12^2} + \frac{1.50(1.25)^2(1.05)}{(0.12 - 0.05) \cdot 1.12^3}$$

$$= \$1.67 + \$1.87 + \$25.03 = \$28.57$$

B is incorrect. The terminal value is not discounted to the present.

$$V = \frac{1.50(1.25)}{1.12} + \frac{1.50(1.25)^2}{1.12^2} + \frac{1.50(1.25)^2(1.05)}{(0.12 - 0.05) \cdot 1.12^2}$$

$$= \$1.67 + \$1.87 + \$35.16 = \$38.70$$

Equity Valuation: Concepts and Basic Tools

82 An investor wants to determine the intrinsic value of the common stock for a company with the following characteristics:

- The firm maintains a constant dividend payout ratio.
- Goodwill and patents account for a high proportion of the firm's assets.
- The firm's revenues and earnings are highly correlated with the business cycle.

Furthermore, the investor focuses on the firm's capacity to pay dividends rather than expected dividends. Considering the characteristics, the investor will *most likely* use which of the following valuation models?

- A** Free cash flow to equity model
- B** Gordon dividend growth model
- C** Asset-based valuation model

A is correct. The free cash flow to equity (FCFE) model is a measure of the firm's dividend-paying capacity, which should be reflected in the cash flow estimates rather than expected dividends. Analysts must make projections of financials to forecast future FCFE, and thus the constant growth assumption, as in the Gordon growth model, is not an issue. An asset-based valuation model is not appropriate because of the high proportion of intangibles (goodwill and patents) in the firm's assets.

B is incorrect. The Gordon model is not appropriate because the investor focuses on the firm's capacity to pay dividends rather than expected dividends. Further, the cyclical nature of the firm's earnings and the constant payout ratio are not consistent with the constant growth rate of dividends assumed under the Gordon model.

C is incorrect. An asset-based valuation model is not appropriate considering the high proportion of intangibles (goodwill and patents) in the firm's assets.

Equity Valuation: Concepts and Basic Tools

- 83** An investor considering the enterprise value approach to valuation gathers the following data:

Earnings before interest, taxes, depreciation, and amortization (EBITDA)	\$65.8 million
Value of debt	\$90.0 million
Value of preferred stock	\$25.4 million
Cash and marketable securities	\$6.9 million
Number of common shares outstanding	12.5 million
Firm's tax rate	30%
EV/EBITDA multiple	6×

The value per share of the company's common stock is *closest* to:

- A** \$13.43.
B \$22.35.
C \$22.90.

C is correct. First, compute the enterprise value (EV) from EBITDA $\times$ EV/EBITDA multiple.

Next, determine market capitalization (value of equity per share) using the following expression:

$$\text{EV} = \text{Market capitalization} + \text{Market value (MV) of preferred stock} + \text{MV of debt} - \text{Cash and investments}$$

$$\text{Market capitalization} = \text{EV} - \text{MV of preferred stock} - \text{MV of debt} + \text{Cash and investments}$$

$$\text{Value per share} = \text{Market capitalization} / \text{Number of outstanding shares}$$

Enterprise value = 65.8×6	394.8
– Value of debt	–90.0
– Value of preferred stock	–25.4
+ Cash and marketable securities	6.9
= Market capitalization, or value of equity	286.3
Value per share = $286.3 / 12.5$	\$22.90

A is incorrect. It adjusts EBITDA for tax effect.

Enterprise value = $65.8 \times (1 - 0.30) \times 6$	276.4
– Value of debt	–90.0
– Value of preferred stock	–25.4
+ Cash and marketable securities	6.9
= Market capitalization or Value of equity	167.9
Value per share = $167.9 / 12.5$	\$13.43

B is incorrect. It ignores adjusting for cash and marketable securities.

Enterprise value = 65.8×6	394.8
– Value of debt	–90.0
– Value of preferred stock	–25.4
+ Cash and marketable securities	N/A
= Market capitalization or Value of equity	279.40
Value per share = $279.40/12.5$	\$22.35

Equity Valuation: Concepts and Basic Tools

- 84 Which of the following situations will *most likely* indicate a reduction of market efficiency?
- A An increase in rules and regulations that promote financial disclosure
 - B An increase in arbitrage opportunities
 - C An increase in the number of market participants

B is correct. Arbitrage is a set of transactions that produces riskless profits. Arbitrageurs are traders who engage in such trades to benefit from pricing discrepancies (inefficiencies) in markets. Such trading activity contributes to market efficiency. If arbitrage opportunities increase, it means that there are either more pricing discrepancies or fewer arbitrageurs (or both), than before indicating that there has been a reduction of market efficiency.

A is incorrect. An increase of rules and regulations that promote financial disclosure will lead to an increase of market efficiency.

C is incorrect. An increase of the number of market participants will lead to an increase market efficiency.

Market Efficiency

- 85 If markets are only weak-form efficient, which of the following investment approaches is *least likely* to consistently earn abnormal profits?
- A Exploiting of non-public information
 - B Buying and selling based on fundamental analysis
 - C Trading based on patterns of prices and volume

C is correct. In the weak-form efficient market hypothesis, security prices fully reflect all past market data, which refers to all historical price and trading volume information. Consequently, trading by looking at patterns of prices and volume from the past cannot consistently earn abnormal profits.

B is incorrect. Fundamental analysis is the examination of publicly available information and the formulation of forecasts to estimate the intrinsic value of assets. If markets are weak-form efficient, only past market data, but not all publicly available information, are reflected in current prices. Consequently, buying and selling based on fundamental analysis can consistently earn abnormal profits.

A is incorrect. If markets are weak-form efficient, only past market data, but not private (nonpublic) information, are reflected in current prices. Consequently, exploiting nonpublic information can consistently earn abnormal profits. In most countries, such activity is prohibited.

Market Efficiency

- 86** Present value models follow a fundamental tenet of economics that states that individuals invest:
- A** to defer consumption.
 - B** based on the law of one price.
 - C** for the expected future benefits.

C is correct. Present value models follow a fundamental tenet of economics stating that individuals defer consumption—that is, they invest—for the future benefits expected.

A is incorrect because, while people must defer consumption to invest, their motivation for doing so is that they expect a rate of return over the investment period that will create future benefits.

B is incorrect because, while the law of one price is the economic rationale underlying the method of comparables for judging asset valuation, it is not the economic reasoning that explains why people invest.

Equity Valuation: Concepts and Basic Tools

- 87** Which of the following is *least likely* to be a type of embedded option in a bond issue granted to bondholders? The right to:
- A** put the issue.
 - B** convert the issue.
 - C** call the issue.

C is correct. The right to call an issue is a type of embedded option granted to issuers, not bondholders. The other two rights are embedded options granted to bondholders.

A is incorrect because this is a type of embedded option granted to bondholders.

B is incorrect because this is a type of embedded option granted to bondholders.

Fixed-Income Securities: Defining Elements

- 88** An investor purchases a 5% coupon bond maturing in 15 years for par value. Immediately after purchase, the yield required by the market increases. The investor would then *most likely* have to sell the bond at:
- A** a premium.
 - B** a discount.
 - C** par.

B is correct. The bond would sell below par or at a discount if the yield required by the market rises above the coupon rate. Because the bond initially was purchased at par, the coupon rate equals the yield required by the market. Subsequently, if yields rise above the coupon, the bond's market price would fall below par.

A is incorrect because the yield is now greater than the coupon.

C is incorrect because the yield is now greater than the coupon.

Introduction to Fixed-Income Valuation

- 89 A 6% 25-year bond with semiannual payments has a market price of \$850.00. The yield to maturity of this bond is *closest to*:

- A 7.32%.
- B 7.91%.
- C 5.72%.

A is correct. The yield to maturity is the discount rate that equates the price of the bond (\$850.00) with its cash flows (49 semiannual cash flows of \$30 and a 50th cash flow of \$1,030), or

$$\$850 = \frac{\$30}{(1 + \text{YTM}/2)^1} + \frac{\$30}{(1 + \text{YTM}/2)^2} + \dots + \frac{\$30}{(1 + \text{YTM}/2)^{49}} + \frac{\$1,030}{(1 + \text{YTM}/2)^{50}}$$

Using a financial calculator to find the yield to maturity gives 7.32%.

B is incorrect because this is the YTM found when setting a calculator for 2 payments per year but inputting only 25 payments, rather than 50.

C is incorrect because this is the YTM found when reversing the price and face value.

Introduction to Fixed-Income Valuation

- 90 Which of the following is *most likely* an example of a Eurobond?

- A A Canadian borrower issuing British pound–denominated bonds in the UK market.
- B A Japanese borrower issuing US dollar–denominated bonds in the US market.
- C An Australian borrower issuing Canadian dollar–denominated bonds in the UK market.

C is correct. A Eurobond is an international bond issued outside the jurisdiction of any one country and not denominated in the currency of the country where it is issued.

A is incorrect because this is an example of a foreign bond, which is a bond issued by an entity incorporated in a country other than where the bond is issued and in whose currency the bond is denominated.

B is incorrect because this is an example of a foreign bond, which is a bond issued by an entity incorporated in a country other than where the bond is issued and in whose currency the bond is denominated.

Fixed-Income Securities: Defining Elements

- 91 Which of the following are *most likely* a kind of supranational bonds? Bonds issued by the:
- A Federal Farm Agency of the United States.
 - B Government of Malaysia.
 - C European Investment Bank.

C is correct. Supranational bonds are bonds issued by such supranational agencies as the European Investment Bank and the International Monetary Fund.

A is incorrect because bonds issued by Federal Farm Agency of the United States are a type of quasi-government bonds.

B is incorrect because bonds issued by the government of Malaysia are a type of government bonds.

Fixed-Income Markets: Issuance, Trading, and Funding

- 92 During the lockout period for a non-amortizing asset-backed security, the principal payment of €100 million on a €1 billion face value issue will result in the security having a total face value of:
- A €0.9 billion.
 - B €1.1 billion.
 - C €1.0 billion.

C is correct. During the lockout period any principal received is reinvested to acquire additional loans with a principal equal to the total principal received from the cash flow keeping the face value of the issue at €1 billion.

A is incorrect because principal received during the lockout period is reinvested in additional loans, and not used to pay down the outstanding issuance.

B is incorrect because the payment of €100 million would not result in an increase in the face value of the issuance, regardless of whether the issue was inside or outside of the lockout period.

Introduction to Asset-Backed Securities

- 93 The weighted average number of years to receipt of the principal and interest payments that will result in realization of the initial market discount rate on a bond is *best* described as:
- A effective duration.
 - B modified duration.
 - C Macaulay duration.

C is correct. Macaulay duration is a weighted average of the time to receipt (expressed in years) of a bond's promised payments where the weights are the shares of the full price that correspond to each of the bond's promised future payments. This weighted average results in the realization of the bond's initial market discount rate.

A is incorrect because effective duration is a measure of the sensitivity of the bond's price (expressed as a percentage change) to a change in the benchmark yield curve.

B is incorrect because modified duration provides an estimate of a bond's percentage price change given a change in its yield to maturity (rather than a measure expressed as a weighted average period of years).

Understanding Fixed-Income Risk and Return

94 Consider the following information relating to a corporate bond:

Full price of bond (PV^{Full})	\$100,367,242
Modified duration (AnnModDur)	8.124 years
Accrued interest	\$303,218

The money duration of the corporate bond is *closest* to:

- A \$812,920,131.
- B \$815,383,474.
- C \$817,846,817.

B is correct. The money duration of the corporate bond is equal to \$815,383,474.

$$\begin{aligned}\text{MoneyDur} &= \text{AnnModDur} \times PV^{\text{Full}} \\ &= 8.124 \times \$100,367,242 \\ &= \$815,383,474\end{aligned}$$

A is incorrect because the full price of the bond (PV^{Full}) includes accrued interest. The calculation of money duration should include accrued interest. However, accrued interest is incorrectly subtracted from the full price of the bond here:

$$\begin{aligned}\text{MoneyDur} &= \text{AnnModDur} \times [PV^{\text{Full}} - \text{Accrued interest}] \\ &= 8.124 \times [\$100,367,242 - \$303,218] \\ &= \$812,920,131\end{aligned}$$

C is incorrect because the full price of the bond (PV^{Full}) already includes accrued interest. Accrued interest was incorrectly added to the full price of the bond here:

$$\begin{aligned}\text{MoneyDur} &= \text{AnnModDur} \times [PV^{\text{Full}} + \text{Accrued interest}] \\ &= 8.124 \times [\$100,367,242 + \$303,218] \\ &= \$817,846,817\end{aligned}$$

Understanding Fixed-Income Risk and Return

95 The table below shows the debt structures of three companies with identical corporate family ratings:

	Company A	Company B	Company C
Senior secured	40.00%	60.00%	40.00%
Senior unsecured	30.00%	30.00%	40.00%
Senior subordinated	15.00%	5.00%	10.00%
Subordinated	15.00%	5.00%	10.00%

Which company will *most likely* exhibit the largest notching adjustment?

- A Company A
- B Company B
- C Company C

B is correct. Generally, the lower the senior unsecured rating, the larger the notching adjustment. In this case, the three companies have identical corporate family ratings, so we would next analyze each company's debt structure to determine which company's issues would most likely exhibit the greatest loss severity given default. Company B has substantially more secured debt than either Company A or Company C in its debt structure, so Company B's lower-ranked issues will likely exhibit lower recovery rates and require larger notching adjustments than either Company A or Company C.

A is incorrect. Notwithstanding the fact that Company A has the largest percentage of subordinated debt in its capital structure, Company A would most likely experience a smaller notching adjustment than Company B because there is a lower percentage of higher-priority claims (secured debt) in its capital structure, which should result in higher recovery rates for its lower-rated credits (subordinated bonds). Higher recovery rates for subordinate bonds translate into smaller notching adjustments.

C is incorrect. Company C would most likely experience a smaller notching adjustment than Company B because there is a lower percentage of higher-priority claims (secured debt) in its capital structure, which should result in higher recovery rates for lower-rated credits (subordinated bonds). Higher recovery rates for subordinated bonds translate into smaller notching adjustments.

Fundamentals of Credit Analysis

96 Selected data for three companies are provided in the table below:

	Company A	Company B	Company C
Total Debt (\$m)	1,125	1,360	1,562
EBITDA (\$m)	590	680	750
Interest Expense (\$m)	71	60	63

Which company's leverage contributes *most adversely* to its credit risk?

- A Company A
- B Company B
- C Company C

C is correct. The only leverage ratio that can be calculated given the data provided is the debt-to-EBITDA ratio. Company C has the highest debt-to-EBITDA ratio of the three companies presented. A higher ratio indicates more leverage and thus higher credit

risk. Company C also has the highest coverage ratio (EBITDA/interest expense) of the three companies, which would indicate lower credit risk, but the question is confined to analyzing leverage ratios only.

Company A: $1,125/590 = 1.9\times$

Company B: $1,360/680 = 2.0\times$

Company C: $1,562/750 = 2.08\times$

A is incorrect. Company A has a debt-to-EBITDA ratio of 1.9, which is lower than Company C's ratio of 2.08. A lower ratio indicates less leverage and less credit risk. (Section 5.2.1, p. 616)

Company A: $1,125/590 = 1.9\times$

Company B: $1,360/680 = 2.0\times$

Company C: $1,562/750 = 2.08\times$

B is incorrect. Company B has a debt-to-EBITDA ratio of 2.0, which is slightly less than Company C's ratio of 2.08. A lower ratio indicates less leverage and less credit risk. (Section 5.2.1, p. 616)

Company A: $1,125/590 = 1.9\times$

Company B: $1,360/680 = 2.0\times$

Company C: $1,562/750 = 2.08\times$

Fundamentals of Credit Analysis

97 As interest rates rise and fall, investors in mortgaged-backed securities *most likely* face what type of risk?

- A** Extension risk and contraction risk
- B** Single-month mortality (SMM) risk and contraction risk
- C** Conditional prepayment rate (CPR) risk and extension risk



A is correct. Extension risk (contraction risk) is the risk that when interest rates rise (decline), actual prepayments will be lower (higher) than forecasted. At lower rates, homeowners will refinance at the now-available lower interest rates. Thus, a security backed by mortgages will have a shorter maturity than was anticipated at the time of purchase. At higher rates, homeowners are reluctant to give up the benefits of a contractual interest rate that now looks low, thus slowing the prepayment on the securitization, leading to its longer maturity. SMM is a measure of prepayment and not a risk, and CPR is the corresponding annualized rate.

B is incorrect. SMM is a measure of prepayment and not a risk.

C is incorrect. CPR is the corresponding annualized rate for SMM.

Introduction to Asset-Backed Securities

98 A sovereign bond will *most likely* be:

- A** issued in the local currency.
- B** backed by the issuer's taxing authority.
- C** virtually free of credit risk when issued.

B is correct. The term *sovereign bond* refers to a debt security issued by a national government with taxing authority. These bonds are typically unsecured and backed by the government's ability to tax.

A is incorrect. A sovereign might issue a bond in a foreign currency if the local currency is relatively illiquid or if there is relatively little demand for the local currency among global investors.

C is incorrect. The credit risk of the issue will depend on the sovereign rating, which, in turn, depends on the issuer's budget surplus/deficit, the currency in which the bonds are issued, and so on. Only issues rated AAA/Aaa would be viewed as virtually credit risk free.

Fixed-Income Markets: Issuance, Trading, and Funding

99 An analyst gathers the following information on a bond:

Price	85.4734
Payment frequency	Annual
Time to maturity	5 years
Coupon	4%
Macaulay duration	4.5947

Modified duration is *closest* to:

- A 4.2702.
- B 4.4180.
- C 4.7785.

A is correct. Modified duration is Macaulay duration divided by 1 plus yield per period. Macaulay duration is provided in the table, so YTM is calculated as follows:

$$N = 5.$$

$$PV = 85.4734.$$

$$PMT = -4.$$

$$FV = -100.$$

$$CPT I/Y = 7.6\%.$$

$$\text{ModDur} = \frac{4.5947}{1.076} = 4.2702$$

B is incorrect. It is the result of incorrectly dividing Macaulay duration by the coupon rate, rather than YTM.

C is incorrect. It is the result of incorrectly multiplying Macaulay duration by 1 plus the coupon rate.

Understanding Fixed-Income Risk and Return

100 According to put–call parity, if a fiduciary call expires in the money, the payoff is *most likely* equal to the:

- A difference between the market value of the asset and the face value of the risk-free bond.
- B market value of the asset.

- C face value of the risk-free bond.

B is correct. A fiduciary call, defined as a long position in a call and in a risk-free bond, generates a payoff that is equal to the market value of the asset if it expires in the money.

A is incorrect. The difference between the market value of the asset and the face value of the risk-free bond is the payoff of the long call if exercised. This ignores the fact that the face value of the bond needs to be added to the payoff.

C is incorrect. The face value of the risk-free bond is the payoff of the fiduciary call if the call expires out of the money,

Basics of Derivative Pricing and Valuation

- 101** A corporation issues five-year fixed-rate bonds. Its treasurer expects interest rates to decline for all maturities for at least the next year. She enters into a one-year agreement with a bank to receive quarterly fixed-rate payments and to make payments based on floating rates benchmarked on three-month Libor. This agreement is *best* described as a:
- A futures contract.
 - B forward contract.
 - C swap.

C is correct. A swap is a series of forward payments. Specifically, a swap is an agreement between two parties to exchange a series of future cash flows. The corporation receives fixed interest rate payments and makes variable interest rate payments. Given that the contract is for one year and the floating rate is based on three-month Libor, at least four payments will be made during the year.

A is incorrect. A forward contract includes one payment only. The swap described has a series of four quarterly payments.

B is incorrect. The instrument described is a swap.

Derivative Markets and Instruments

- 102** During its life, the value of a forward contract is *most likely* equal to the price of the underlying minus the price of the:
- A forward.
 - B forward, discounted over the original term of the contract.
 - C forward, discounted over the remaining term of the contract.

C is correct. The value of a forward contract is the spot price of the underlying minus the present value of the forward contract. Calculating the present value requires adjusting the time period to account for the remaining term of the contract.

A is incorrect. This is only true at expiration.

B is incorrect. This is only true at initiation.

Basics of Derivative Pricing and Valuation

- 103** Which statement best describes the early exercise of non-dividend paying American options? Early exercise may be advantageous for:
- A** both deep-in-the-money calls and deep-in-the-money puts.
 - B** deep-in-the-money calls.
 - C** deep-in-the-money puts.

C is correct. Only deep-in-the-money put options may be exercised early. The price cannot fall below zero, so the additional upside of such an option is limited.

A is incorrect. Being deep in the money is no reason for an early exercise of call options because there are no theoretical limits to further price increases.

B is incorrect. Being deep in the money is no reason for an early exercise of call options because there are no theoretical limits to further price increases.

Basics of Derivative Pricing and Valuation

- 104** What is the *most likely* reason why arbitrage will not completely eliminate all pricing discrepancies for derivatives?
- A** Differences in risk aversion
 - B** Transaction costs
 - C** Inaccurate forecasts

B is correct. Transaction costs may render an arbitrage strategy unprofitable and can therefore prevent precise convergence of prices.

A is incorrect. Differences in risk aversion are irrelevant for arbitrage because arbitrage transactions are riskless.

C is incorrect. No forecasts are needed in implementing an arbitrage position.

Basics of Derivative Pricing and Valuation

- 105** Relative to spot markets, one key feature of derivatives markets is:
- A** high transaction costs.
 - B** low capital requirements.
 - C** restrictions on short selling.

B is correct. Derivatives markets provide the benefit of low capital requirements to hedgers and speculators.

A is incorrect. Derivatives markets provide the benefit of low transaction costs to hedgers and speculators.

C is incorrect. Derivatives markets provide the benefit of ease of short selling to hedgers and speculators.

Derivative Markets and Instruments

106 A European call option is in the money whenever the value of the underlying is:

- A** equal to the exercise price.
- B** less than the exercise price.
- C** greater than the exercise price.

C is correct. A European call option is in the money when the value of the underlying is greater than the exercise price of the option.

A is incorrect. A European call option is at the money when the value of the underlying is equal to the exercise price of the option.

B is incorrect. A European call option is out of the money when the value of the underlying is less than the exercise price of the option.

Basics of Derivative Pricing and Valuation

107 If an investor uses derivatives to make a long investment in commodities, the return earned on margin is *best* described as:

- A** convenience yield.
- B** collateral yield.
- C** price return.

B is correct. Collateral yield is the return on cash used as margin on derivatives used to gain commodity exposure.

A is incorrect because the convenience yield (also known as "roll yield") is the return from rolling forward the maturity of the derivatives position.

C is incorrect because price return is the difference between the forward and spot price.

Introduction to Alternative Investments

108 Do management fees *most likely* get paid to the manager of a hedge fund, regardless of the fund's performance?

- A** No, only when the fund's net asset value exceeds the previous high-water mark
- B** No, only when the fund's gross return is positive
- C** Yes

C is correct. Regardless of performance, the management fee is always paid to the fund manager.

B is incorrect because the gross return can be at any level and the manager is still paid the management fee.

A is incorrect because the management fee is paid regardless of the value of the assets in the fund.

Introduction to Alternative Investments

109 Concentrated portfolio strategies are attractive because of their:

- A** potential to generate alpha.
- B** ability to track market indices.
- C** low risk.

A is correct. Concentrated portfolio strategies focus on only a few securities, strategies, or managers. This focus reduces diversification but may enable investors to achieve alpha.

B is incorrect. Portfolio concentration makes it harder to track market indexes.

C is incorrect. Portfolio concentration increases risk.

Introduction to Alternative Investments

110 A commodity market is in contango when futures prices are:

- A** lower than the spot price.
- B** the same as the spot price.
- C** higher than the spot price.

C is correct. When a commodity market is in contango, futures prices are higher than the spot price.

A is incorrect. This is the definition of backwardation.

B is incorrect. This is neither contango nor backwardation.

Introduction to Alternative Investments

111 Which of the following is *least likely* to reduce the likelihood of being defrauded by a dishonest money manager?

- A** Third-party custody of assets under management
- B** Strong and consistent reported investment performance
- C** Independent verification of investment results

B is correct. To prevent fraud, involvement of third parties in the reporting and asset management process is helpful. A strong and consistent reported investment performance that lacks outside verification may actually be a warning sign.

A is incorrect. Third-party custody of assets under management helps to reduce the possibility of fraud.

C is incorrect. Independent verification of investment results helps to reduce the possibility of fraud.

Introduction to Alternative Investments

112 A private equity firm sells a portfolio company to a buyer that is active in the same industry as the portfolio company. This transaction is best described as a(n):

- A** trade sale.
- B** secondary sale.
- C** initial public offering.



A is correct. A trade sale is the sale of a portfolio company to a strategic buyer, such as a company that is active in the same industry.

B is incorrect. A secondary sale is a sale to another private equity firm.

C is incorrect. An initial public offering involves the sale of shares to public investors.

Introduction to Alternative Investments

113 The market approach to valuing portfolio companies in private equity firms is *most likely* based on:

- A** present value.
- B** the value of assets minus the value of liabilities.
- C** multiples.



C is correct. The market approach to valuing portfolio companies uses multiples of different measures that are compared with similar companies.

A is incorrect. Present value is calculated in the context of discounted cash flow models.

B is incorrect. The value of assets minus the value of liabilities is calculated in the context of asset-based models.

Introduction to Alternative Investments

114 The relative strength index for a stock stands at 75. This reading is *best* described as an indication that the stock is

- A** neutral.
- B** oversold.
- C** overbought.



C is correct. The relative strength index (RSI) is a momentum oscillator and provides information on whether or not an asset is overbought or oversold. An RSI greater than 70 indicates that a stock is overbought; an RSI lower than 30 suggests that a stock is oversold.

A is incorrect. The relative strength index is a momentum oscillator and provides information on whether or not an asset is overbought or oversold. An RSI greater than 70 indicates that a stock is overbought; an RSI lower than 30 suggests that a stock is oversold.

B is incorrect. The relative strength index is a momentum oscillator and provides information on whether or not an asset is overbought or oversold. An RSI greater than 70 indicates that a stock is overbought; an RSI lower than 30 suggests that a stock is oversold.

Technical Analysis

115 Which of the following institutional investors is *most likely* to have a low tolerance for investment risk and relatively high liquidity needs?

- A** Insurance company
- B** Defined-benefit pension plan
- C** Charitable foundation

A is correct. Insurance companies need to be relatively conservative and liquid, given the necessity of paying claims when due.

B is incorrect because defined-benefit pension plans tend to have quite high risk tolerances and quite low liquidity needs.

C is incorrect because endowments/foundations typically have high risk tolerances and quite low liquidity needs.

Portfolio Management: An Overview

116 The risk-free rate is 5%, and the market risk premium is 8%. If the beta of TRL Corp. is 1.5, based on the capital asset pricing model (CAPM), the expected return of TRL's stock is *closest* to:

- A** 17.0%.
- B** 9.5%.
- C** 15.5%.

A is correct. Using the CAPM relationship of $E(R_i) = R_f + [E(R_m) - R_f]\beta_i$, we can estimate the expected return as: $E(R_i) = 0.05 + (0.08)(1.5) = 17.0\%$.

B is incorrect because the expected return is computed as $0.05 + (0.08 - 0.05)(1.5) = 9.5\%$.

C is incorrect because the expected return is computed as $0.08 + (0.05)(1.5) = 15.5\%$.

Portfolio Risk and Return: Part II

117 An investment policy statement's risk objective states that over a 12-month period, with a probability of 95%, the client's portfolio must not lose more than 5% of its value. This statement is *most likely* a(n):

- A** total risk objective.
- B** relative risk objective.
- C** absolute risk objective.



C is correct. The statement is an absolute risk objective because it expresses a maximum loss in value with an associated probability of loss.

A is incorrect because this is an absolute (not total) risk objective because it expresses a maximum loss in value with an associated probability of loss.

B is incorrect because this is an absolute (not relative) risk objective because it expresses a maximum loss in value with an associated probability of loss.

Basics of Portfolio Planning and Construction

118 Two investors have utility functions that differ only with regard to the coefficient of risk aversion. Relative to the investor with a higher coefficient of risk aversion, the optimal portfolio for the investor with a lower coefficient of risk aversion will *most likely* have:

- A** a lower level of risk and return.
- B** a higher level of risk and return.
- C** the same level of risk and return.



B is correct. A less risk-averse investor's highest utility curve, given the lower coefficient of risk aversion, is likely to touch the capital allocation line at a point that would represent a portfolio with higher risk and more expected return.

A is incorrect because for a high coefficient of risk aversion, the investor will seek a lot of return for a bit of extra risk and will have an optimal portfolio that is tangential to the capital allocation line at a lower level of risk and return relative to the investor with a lower coefficient of risk aversion.

C is incorrect because only investors with identical coefficients of risk aversion would select the same optimal portfolio.

Portfolio Risk and Return: Part I

119 Within a risk management framework, risk tolerance:

- A** and risk exposure should be kept in alignment.
- B** includes the qualitative assessment and evaluation of risk.
- C** is determined as a result of establishing how and where risk is taken.



A is correct. The process of risk monitoring, mitigation, and management is the most obvious facet of the risk management framework and requires recognizing when risk exposure is not aligned with risk tolerance.

B is incorrect. Risk identification and measurement include the qualitative assessment and evaluation of risk.

C is incorrect. Risk tolerance can provide guidance on risk budgeting, which is how and where risk is taken. Risk budgeting is a result of determining risk tolerance.

Risk Management: An Introduction

- 120 Which of the following statements *best* describes a potential concern for clients using robo-advisers? Robo-advisers:
- A must be established as registered investment advisers.
 - B do not seem to incorporate the full range of investment information into their recommendations.
 - C are likely to be held to a similar code of conduct as other investment professionals in the given region.



B is correct. Initial research has shown that robo-advisers do not seem to incorporate the full range of investment information into their recommendations, meaning that important points may be missing in investment decisions.

A is incorrect. It is an advantage to clients that robo-advisers must be registered as investment advisers, because they are subject to guidance from the securities regulator of their country.

C is incorrect. It is an advantage to clients that robo-advisers must be held to a similar code of conduct as other investment professionals in the given region.

Fintech in Investment Management